

Engineering Standard

02 May 2023

SAES-X-400 Cathodic Protection of Buried Pipelines

Document Responsibility: Cathodic Protection Standards Committee

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Summary of Changes

Paragraph Number		Change Type (Addition, Modification, Deletion, New)	Technical Change(s)
Previous Revision 01 February 2021	Current Revision 02 May 2023		
5.1	5.1	Addition	Requirement for design package and construction drawings related to Alternating Current Mitigation (ACM) added.
6.10.1.4	6.10.1.4	Modification	Bond wire size should be based on the bond current and attenuation.
6.13.1	6.13.1	Modification	Material specifications for RMU shall be as per 17-SAMSS-018
6.13.1.4	6.12.2.2	Addition	Remote monitoring units shall be provided along the length of pipelines at every 5 km intervals and at critical locations listed
6.13.2.2	6.13.2.2	Modification	Provide a four-wire current measurement test stations requirement detailed
	6.15	Addition	AC Interference study, modelling and mitigation and monitoring requirement has been detailed
7.1	7.1	Modification	Note-1 modified
7.1	7.1	Modification	Note-3, Current density values of 0.2 mA/m ² for Wadi & Sabkha.
	8	Addition	added AC Mitigation System Equipment and Material Requirements
	8.2.4	Addition	Decision to select Photovoltaic Power (PV) as per economic evaluation
8.5	8.5	Modification	Use of other type of CP assessment probes and or coupons are permitted as selected by the proponent

Paragraph Number		Change Type (Addition, Modification, Deletion, New)	Technical Change(s)
Previous Revision 01 February 2021	Current Revision 02 May 2023		
	8.7	Addition	Selection and use of DC Decouplers & Zinc Grounding for AC mitigation added

1 Scope

- 1.1** This standard prescribes the minimum mandatory requirements governing the design and installation of cathodic protection systems for onshore pressurized buried metallic pipelines located outside of plant facilities. Typical pipelines addressed by this standard are; cross-country, production, and utility pipelines.
- 1.2** Subsea pipelines and pipelines operated by Saudi Aramco Community Maintenance Departments are outside the scope of this standard (refer to SAES-X-300 for cathodic protection of marine structures).

2 Conflicts and Deviations

Any conflicts between this document and other applicable Mandatory Saudi Aramco Engineering Requirements (MSAERs) shall be addressed to the EK&RD Manager.

Any deviation from the requirements herein shall follow internal company procedure SAEP-302.

3 References

All referenced specifications, standards, codes, drawings, and similar material are considered part of this Engineering Standard to the extent specified, applying the latest version, unless otherwise stated.

3.1 Saudi Aramco References

Saudi Aramco Engineering Procedures

SAEP-302	Waiver of a Mandatory Saudi Aramco Engineering Requirement
SAEP-303	Engineering Reviews of Project Proposal and Detail Design Documentation
SAEP-332	Cathodic Protection Commissioning
SAEP-333	Cathodic Protection Monitoring

Saudi Aramco Materials System Specification

17-SAMSS-004	Conventional Rectifiers for Cathodic Protection
17-SAMSS-006	Galvanic Anodes for Cathodic Protection
17-SAMSS-007	Impressed Current Anodes for Cathodic Protection
17-SAMSS-008	Junction Boxes for Cathodic Protection
17-SAMSS-012	Photovoltaic Power Supply for Cathodic Protection
17-SAMSS-017	Impressed Current Cathodic Protection Cables
17-SAMSS-018	Remote Monitoring Units (RMUs) for Cathodic Protection

Saudi Aramco Engineering Standards

SAES-B-064	Onshore and Nearshore Pipeline Safety
SAES-H-002	Internal and External Coatings for Steel Pipelines and Piping
SAES-L-460	Pipeline Crossings under Roads and Railroads
SAES-P-111	Grounding
SAES-X-300	Cathodic Protection of Marine Structures
SAES-X-600	Cathodic Protection of External Surfaces of Plant Facilities
SAES-X-700	Cathodic Protection of Onshore Well Casings
SAES-A-134	External Corrosion Protection Requirements

Saudi Aramco Best Practices

SABP-X-001	CP Design Package Preparation
SABP-X-003	Cathodic Protection Installation Requirements

Saudi Aramco Standard Drawings

AA-036069	Galvanic Anodes at Thrust Anchors
AD-036132	Termination Detail Cable Identification
AA-036145	Cable Splice Junction Box
AA-036157	Galvanic Anode 3-Pin Test Station
AB-036273	Surface Marker Underground Electric Cable
AB-036274	5 Terminal Junction Box
AB-036275	12 Terminal Junction Box
AA-036276	Multi-Purpose Junction Box
AA-036277	5 Terminal Bond Box Details
AA-036346	Surface Anode Bed Details - Horizontal and Vertical Anodes (Sheets 1 & 2)
AA-036347	20 Terminal Junction Box
AA-036349	3 Terminal Bond Box Details
AA-036350	2 Terminal Bond Box Details
AB-036351	Marker Plate Details
AA-036352	Galvanic Anodes for Road and Camel Pipeline Crossings (Sheets 1 & 2)
AB-036378	Rectifier Installation Details (Sheets 1 & 2)
AB-036381	Cable Connections to Pipelines and Structures
AA-036385	Deep Anode Bed without Anode Support Pipe
AA-036389	Galvanic Anode Details
AB-036540	Mounting Support Details for Junction Boxes
AA-036674	Bonding Details for Onshore Pipelines and Flowlines
AA-036675	Direct Buried Cable - Installation Details
AD-036785	Symbols - Cathodic Protection
AB-036907	Test Stations for Buried Pipelines

Saudi Aramco General Instructions

GI-0002.710	Mechanical Completion and Performance Acceptance of Facilities
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3.2

Other References

National Fire Protection Association

NFPA/NEC Handbook	National Fire Protection Association Publication of the National Electrical Code
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The Association for Materials Protection and Performance (AMPP/NACE)

AMPP SP0104 – 2020	The Use of Coupons for Cathodic Protection Monitoring Applications
AMPP SP0169 - 2013	Control of External Corrosion on Underground or Submerged Metallic Piping Systems
AMPP SP0177 - 2019	Mitigation of Alternating Current and Lightning Effects on Metallic Structures and Corrosion Control Systems

AMPP SP21424-2018	Alternating Current Corrosion on Cathodically Protected Pipelines
AMPP SP0200 - 2014	Steel-Cased Pipeline Practices
AMPP SP0286 - 2007	Electrical Isolation of Cathodically Protected Pipelines
AMPP TM0497 - 2022	Measurement Techniques Related to Criteria for Cathodic Protection on Underground or Submerged Metallic Piping Systems

The British Standards Institute and the European Committee for Standards

BS EN 50443 (2011)	Effects of Electromagnetic Interference on Pipelines Caused by High Voltage A.C. Electric Traction Systems and/or High Voltage A.C. Power Supply Systems
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Institute of Electrical and Electronics Engineers

IEEE C57.12	Standard General Requirements for Liquid-Immersed Distribution, Power, and Regulating Transformers
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International Standards Organization

ISO-15589-1 (2015)	Petroleum and Natural Gas Industries - Cathodic Protection of Pipeline Transportation Systems - Part 1: On-land Pipelines
ISO-18086 (2020)	Corrosion of Metals and Alloys - Determination of AC Corrosion Protection criteria
ISO-22426 (2020)	Assessment of the Effectiveness of Cathodic Protection Based on Coupon Measurements

4 Terminology

4.1 Acronyms

AOC	Aramco Overseas Company, The Hague
ASC	Aramco Services Company, Houston
ANSI	American National Standards Institute
ASTM	American Society for Testing and Materials
AWS	American Welding Society
ASME	American Society of Mechanical Engineers (USA)
CP:	Cathodic Protection
CSA	Canadian Standard Association
CSD	The Saudi Aramco Consulting Services Department
FM	Factory Mutual Research Corp.

GOSP	Gas and Oil Separation Plant
HDD:	Horizontal Directional Drilling
ICCP:	Impressed Current Cathodic Protection
JEDEC	Joint Electronic Devices Engineering Council
MSAER	Mandatory Saudi Aramco Engineering Requirements
NEC	National Electrical Code
NEMA	National Electrical Manufacturers Association (USA)
OHPL:	High-voltage overhead or buried power lines.
PMT	Project Management Team used as a truncated version of Saudi Aramco Project Management Team or SAPMT.
RSA	Saudi Aramco Responsible Standardization Agency
SAES	Saudi Aramco Engineering Standard
SAMSS	Saudi Aramco Materials System Specification
SI	International System of Units
SSPC	Steel Structures Painting Council (USA)
UL	Underwriter's Laboratories, Inc. (USA)
WIP:	Water Injection Plant

4.2 Definitions

The following general list of definitions and abbreviations has been developed specifically for application with Saudi Aramco cathodic protection standards and specifications:

AA:	The ANSI cooling class for a dry-type self-cooled transformer or reactor that is cooled by the natural circulation of air. Syn.: AN (IEC), see definition 3.1 in IEEE C57.12.
AC:	Alternating current.
AC Coupon:	An alternating current coupon, with a surface area of 1 cm ² , used to measure representative AC potentials or AC current densities on a pipeline or other buried or immersed metallic structure.
ACM:	Alternating Current Mitigation is the use of a design and construction methodology that specifically reduces the otherwise unsafe AC step and touch potentials on a pipeline to within recognized safe limits. The effects of fault current events and the associated Ground Potential Rise concerns are considered in addition to induced AC voltages arising from Normal and Emergency steady state operating conditions of High Voltage Overhead Powerlines.
Anode:	An impressed current or a galvanic anode for cathodic protection applications.
Anode Cable:	A cable directly connected to an impressed current or galvanic anode.

ANV:	The ANSI cooling class for a dry-type non-ventilated self-cooled transformer, which is so constructed as to provide no intentional circulation of external air through the transformer, and that operates at zero gauge pressure. Syn.: ANAN (IEC), see definition 3.15 in IEEE C57.12.
Armored Cable:	Armored cable for cathodic protection is a single core insulated cable manufactured with a double layer spiral wound insulated galvanized steel sheath to provide mechanical protection, and typically used for subsea applications.
Bond Cable:	A cable installed between two metallic structures to provide electrical continuity between the structures for the purpose of cathodic protection.
Buyer:	Saudi Aramco Purchasing Department Representative.
Buyer's Representative:	The person or persons designated by the Purchasing Department to monitor/enforce the contract. Normally, this is the on-site inspector.
Calcined Petroleum Coke Breeze:	A carbonaceous backfill used as a conductive backfill media for impressed current anodes in soil.
Christmas Tree:	The set of valves, spools and fittings connected to the top of a well to direct and control the flow of formation fluids from the well.
Coated Casing:	The term "coated casing" as used in this engineering standard describes a well casing with an external non-conductive coating (typically Fusion Bonded Epoxy or FBE). The coating must be applied to all sections of the casing in contact with soil or formation, from surface to the bottom of the casing or to a depth determined to facilitate external corrosion mitigation with cathodic protection through the relevant down hole corrosive aquifers. Casings that have been coated over the upper two or three joints of casing only are not "coated casings". Coating applied to well casings is not applied as a corrosion barrier. It is applied to reduce the total amount of CP required or to extend the influence of the applied CP.
CP Assessment Probe:	A multi-electrode probe custom fabricated for Saudi Aramco that is designed to facilitate representative measurements of instant off, total polarization, current density, and resistivity.
CP Coupon:	A cathodic protection coupon used to measure representative potentials or current densities on a pipeline or other buried or immersed metallic structure.

CP System Operating Circuit Resistance:

The total resistance seen at the output of a CP power supply or the total working resistance in a galvanic anode system.

CP System Operating Circuit Resistance:

The total resistance seen at the output of a CP power supply or the total working resistance in a galvanic anode system.

CP System Rated Circuit Resistance:

The rated output voltage of a cathodic protection power supply divided by the rated output current. For photovoltaic power supplies, the rated output current is the design commissioning current.

Cross Country Pipeline:

A pipeline between; two plant areas, another cross country pipeline and a plant area, or between two cross country pipelines.

CSE:

Saturated copper/copper sulfate reference electrode.

CSV:

Coating Stress Voltage. This is the voltage that a pipeline coating will be subjected to in the event of a fault current event arising from a nearby High Voltage Overhead Powerline that may cause coating failure or significant degradation should the dielectric strength of the coating not be high enough to withstand the stress voltage.

Deep Anode Bed:

Anode or anodes connected to a common CP power supply and installed in a vertical hole with a depth exceeding 15 m (50 ft).

DCD:

Direct Current Decoupling Device. This is a device that is installed between the pipeline or structure that may be subject to AC interference with the risk of arcing or high AC potentials developing. These devices may eliminate or greatly reduce the induced potentials resulting during normal operation or surge conditions and reduce the possibility of arcing and structure damage.

Design Agency:

The organization responsible for the design of the pipeline and associated CP system. The Design Agency may be the Design Contractor, the Lump Sum Turn Key Contractor or an in house design organization of Saudi Aramco.

Drain Point:

The location on the cathodically protected structure where the negative cable from the rectifier or negative junction box is fastened to the structure.

EMF:

Electromagnetic field. Magnetic field related to AC interference arising from high voltage AC overhead powerlines.

ERP:

Electrical Resistance Probes and instruments determine metal loss from corrosion or erosion by the electrical resistance method. Electrical resistance instruments directly measure the metal loss on Electrical Resistance Probes.

Flowline:	A pipeline connected to a well.
Flare Line or Blow Down Line:	A line for pumping out of unwanted gas or hydrocarbon
Foundry (for anodes):	An anode foundry is a facility that produces metal castings from either ferrous or non-ferrous metals or alloys.
Galvanic Anodes:	Anodes fabricated from materials such as aluminum, magnesium or zinc that are connected directly to the buried structure to provide cathodic protection current without the requirement for an external cathodic protection power supply. Galvanic anodes are also referred to as sacrificial anodes.
Hazardous Areas:	An area where fire or explosion hazards may exist due to flammable gases or vapors, flammable liquids, combustible dust, or ignitable fibers or filings (see NEC Article 500).
Impressed Current Anodes:	Anodes typically fabricated from High Silicon Cast Iron (HSCI), Mixed Metal Oxide (MMO) or liner anode, etc. that are connected through a DC power supply to the buried or immersed structure to provide cathodic protection current.
Instant-OFF Potential:	OFF potential measured with a short delay after interruption (switching off) of all sources of applied cathodic protection current with the aim of approaching the IR-free potential.
Isolating Spark Gap (ISG):	Component to discharge high voltages from lightning strikes and/or AC fault currents for isolating electrically conductive installation sections to prevent potential damage to isolation devices.
Kill Line:	A high-pressure pipe leading from an outlet on the BOP stack to the high-pressure rig pumps
Lead Wire (for anodes):	A cable directly connected to an anode.
Manufacturer:	The Company that assembles the components for the finished product and provides the finished product either through a Vendor or directly to Saudi Aramco.
Resistivity Meter (Megger):	A meter designed to measure ground resistivity, or can be connected to measure resistance in a format that excludes the resistance of the test wires.
Mill (for anodes):	An anode mill is a facility that produces anodes such as MMO or Platinized Niobium (without lead wires) for cathodic protection applications (or any other anode type that does not involve casting).

Negative Cable:	A cable that is electrically connected (directly or indirectly) to the negative output terminal of a cathodic protection power supply or to a galvanic anode. This includes bond cables to a cathodically protected structure.
Non-metallic Flowline:	A flowline that is made out of non-metallic materials.
Non-incendive Equipment:	Equipment having electrical/electronic circuitry that is incapable, under normal operating conditions, of causing ignition of a specified flammable gas-air, vapor-air, or dust-air mixture due to arcing or thermal means. (See NEC Article 100)
Off-Plot:	Off-plot refers to any area outside of the plot limits.
ON Potential:	Potential measured while all sources of applied cathodic protection current (CP System) is continuously operating.
OFF Potential:	Potential measured after interruption (switching off) of all sources of applied cathodic protection current with the aim of approaching an IR-free potential.
ONAN:	The ANSI cooling class for a transformer or reactor having its core and coils immersed in mineral oil or synthetic insulating liquid with a fire point less than or equal to 300°C, the cooling being effected by the natural circulation of air over the cooling surface. (ONAN was previously termed OA). See definition 3.303 in IEEE C57.12
On-Plot:	On-plot refers to any area inside the plot limit.
Perimeter Fence:	The fence which completely surrounds an area designated by Saudi Aramco for a distinct function.
Photovoltaic Module:	A number of solar cells wired and sealed into an environmentally protected assembly.
Pipeline:	The term "pipeline" is used generically to refer to any type of pipeline.
Plant Area:	The area within the plot limits of a process or storage facility.
Plot Limit:	The plot limit is the boundary around a plant or process facility. The plot limit may be physical such as a fence, a wall, the edge of a road or pipe rack, chains and posts or a boundary indicated on an approved plot plan.
Positive Cable:	A cable that is electrically connected (directly or indirectly) to the positive output terminal of an impressed current cathodic protection power supply, including impressed current anode cables.

Process Pipeline:	A pipeline typically associated with a plant process and typically above ground within a plant facility.
Production Pipeline:	A pipeline transporting oil, gas or water to or from a well. These include flowlines, test lines, water injection lines, and trunklines.
Pyramid Anode:	An offshore anode assembly custom manufactured for Saudi Aramco that is fabricated with mixed metal oxide anode components and placed on the sea bed. The pyramid anode derives its name from the pyramid shaped concrete base.
Reference Electrode:	An industry standardized electrode used as a common reference potential for cathodic protection measurements. A copper/copper sulfate (Cu/CuSO ₄) reference electrode is typically used for soil applications. A silver/silver chloride (Ag/AgCl/0.6M Cl) reference electrode is typically used for aqueous applications.
Responsible Standardization Agent (RSA):	For CP material, this would usually be the Supervisor of the CSD Cathodic Protection Team or the Saudi Aramco CSD cathodic protection Subject Matter Expert.
Soil Transition Point:	The surface location where a pipeline enters or exits the soil, i.e., above grade to below grade transition, or below grade to above grade transition.
Stray Current:	Current flowing through the unintended circuits/paths.
Subject Matter Expert (SME):	The SME is the assigned Consulting Services Department cathodic protection specialist.
Surface Anode Bed:	Anode or anodes connected to a common CP power supply, installed either vertically or horizontally at a depth of less than 15 m (50 ft).
Surge Diverters (SD):	Component to discharge high voltages from lightning strikes and/or AC fault currents for isolating electrically conductive installation sections to prevent potential damage to isolation devices.
Tension Spring Anode:	An offshore anode assembly custom manufactured for Saudi Aramco that is fabricated with mixed metal oxide anode components and suspended on a wire rope/tension spring assembly beneath an offshore platform.
Test Line:	A pipeline that is used for testing an individual well or group of wells.
Thermite Weld:	An exothermic process for use in making electrical connections between two pieces of copper or between copper and steel.

Transformer/Rectifier (T/R):	A CP power supply that transforms and converts AC to DC power, often referred to in the cathodic protection industry as a rectifier.
Transmission Pipeline:	A cross country pipeline transporting product between GOSPs WIPs or other process facilities.
Trunkline:	A pipeline designed to distribute or gather product from two or more wells, typically connecting flowlines or injection lines to the associated GOSP or WIP.
Uncoated Casing:	The term “uncoated casing” when used in this engineering standard describes a well casing that is either bare (no external coating), or may have an external coating applied to the shallow sections of casing to minimize corrosion in the landing base or surface soils.
Unconventional Deep Anode Bed:	Anode bed drilled to typical depth of 120 meters thru dry drilling using air or foam with a geotextile liner to 120 meters and PVC water reservoirs on the top 60 meters.
Utility line:	A pipeline that delivers a service product (typically water, gas or air).
Vendor:	A company that receives a purchase order to supply the finished product, material or equipment. The Vendor may also be the Manufacturer.
Venturi Spool:	A gas metering spool recording apparatus at gas well sites.
Well Casing:	Large-diameter pipe lowered into an open hole and cemented in place.
Well Head:	The system of spools, valves, and assorted adapters that provide pressure control of a production well.

5 Design Review and Approval

- 5.1** The proposed construction drawings and the related cathodic protection and alternating current mitigation (ACM) design package in accordance with SAEP-303 shall be submitted to the CP Proponent organization, proponent’s technical support department as appropriate and to Saudi Aramco’s Consulting Services Department (CSD) for review and approval.

Notes:

- 1) *PMT should request assistance from CSD for the verification of the qualifications of the Design Contractor’s engineer responsible for designing the CP systems.*
- 2) *It is recommended that the CP Design Agency hold a pre-design meeting with CSD, the Proponent CP Group and their technical support department to outline the design approach for the Project Proposal and Detailed Design.*

- 5.2** The Design Agency shall not issue drawings for construction until the drawings have been reviewed and approved in writing by CSD and the CP Proponent organization and their technical support.

5.3 Product hardware and software specifications and relevant design information for all remote monitoring equipment proposed for cathodic protection systems for the Saudi Aramco Pipelines organization shall be submitted to the Supervisor of IT/COD/SCSD (Information Technology/Communication Operations Dept./Satellite & Communication Support Div.) for review and approval during the Project Proposal and Detail Design phases of the Project.

5.4 The Project Proposal package shall provide all general design considerations that can be developed without requiring measurement of field data. The field data including existing CP system and ACM details were available can be provided as an input by the PMT team by collecting the same from Proponent.

Cathodic protection requirements shall be based on the following consideration which shall be clearly specified and agreed with Proponent at the project proposal stage.

- the new pipeline and associated CP equipment to be an independent system
- the new pipeline as a part of the existing pipeline corridor which considers spare capacity of existing systems, existing pipeline protection status and upgradation of existing system

The 90% project proposal package shall be submitted to the CP Proponent organization and their technical support department and to Saudi Aramco's Consulting Services Department (CSD) for review and approval.

The Project Proposal package should include:

- a. A scope of work including a specific statement that clearly identifies any additional requirement to provide CP and ACM for any existing pipeline.
- b. Proposed locations of new CP and ACM systems on an overall CP and ACM system layout drawing, including proposed anode type(s) and estimated output ratings of the proposed cathodic protection power source.
- c. Information on existing cathodic protection system location and spare capacity (from nearby or adjacent pipelines) along the route of the proposed pipeline.
- d. Information on all locations where the proposed pipeline will be mechanically connected to other facilities (plants, pipelines, etc.) with clear details on whether these other facilities will be electrically isolated.

5.5 The 90% Detailed Design package for CP shall contain at minimum:

- a. The completed and signed "Quality Assurance Check List for CP Designs" attached to the cover letter or transmittal sheet for the design package (see Appendix 1).
- b. The scope of work.
- c. One full size professionally drafted Index "X" CP drawing illustrating the entire pipeline length; containing the following details:
 - The proposed location of each piece of CP equipment including but not limited to; rectifiers, anodes, junction boxes, bond stations, test stations, and bonds to structures.
 - Cathodic protection cables including all structure, bond, and rectifier cables; clearly identifying the termination points, cable size, type and length for each cable.
 - All proposed and existing CP installations in the area that may affect the new CP system or pipeline.
 - Existing pipelines that parallel within 50 meters, or cross the proposed pipeline.

- All mechanical tie-ins with details on placement of electrical isolation.
- HV overhead power lines
- Foreign pipeline crossings and parallelism.
- d. All calculations and applicable field data required to verify design compliance with Saudi Aramco Engineering Standards for Cathodic Protection.
- e. All details for the Remote Monitoring System including details on the hardware, software, and connectivity.
- f. The design package shall be submitted in the format specified in Best Practice SABP-X-001 "CP Design Package Preparation."

5.6 The 90% Detailed Design package for ACM shall contain at minimum:

- a. The completed and signed "Quality Assurance Check List for AC Mitigation Designs" attached to the cover letter or transmittal sheet for the design package (see Appendix 2).
- b. The scope of work.
- c. One full size professionally drafted Index "X" ACM drawing illustrating the entire pipeline length; containing the following details:
 - The proposed location of each piece of ACM equipment including but not limited to; DC decoupler devices, earthing anodes, junction boxes, bond stations, test stations, and bonds to structures.
 - ACM cables including all structure, bond, and DC decoupler device cables; clearly identifying the termination points, cable size, type and length for each cable.
 - All proposed and existing ACM installations in the area that may affect or interact with the new ACM system or pipeline.
 - Existing pipelines and powerlines that parallel and cross the proposed pipeline.
 - Nearest powerline tower to the pipeline to be shown on the drawing with the separation distance between the pipeline and tower shown clearly.
 - All mechanical tie-ins with details on placement of electrical isolation.
 - HV overhead power lines crossings and parallelism with details identifying the voltage rating, powerline name and suitable tower numbering.
 - Foreign pipeline crossings and parallelism.
- d. All calculations and applicable field data required to verify design compliance with Saudi Aramco Engineering Standards Cathodic Protection.
- e. All details for the Remote Monitoring System including details on the hardware, software, and connectivity.
- f. All details of the DC decoupler devices, their rating and method of selection.
- g. The design package shall be submitted following a similar format specified in Best Practice SABP-X-001 "CP Design Package Preparation."

6 General Design Requirements

6.1 Contractor/Designer Qualifications and Equipment Requirements

Cathodic protection designs shall be carried out by design offices that are approved by Saudi Aramco. Such design offices must have qualified engineers with a minimum of ten years verifiable cathodic protection design experience and a minimum industry qualification of NACE CP Level 4.

Field measurements required for the design shall be collected by an Engineer or Technician with a minimum of five years verifiable cathodic protection field experience and industry certification level of NACE CP Level 2 or higher.

Installation supervision of cathodic protection materials shall be done by Engineer or Technician with a minimum of five years verifiable cathodic protection field experience and industry certification level of NACE CP Level 2 or higher.

6.2 Fundamental Design Calculations

The detailed design and for a cathodic protection system shall be based on the following consideration which shall be clearly identified in the project proposal.

- the new pipeline and associated CP equipment to be an independent system
- the new pipeline as a part of the existing pipeline corridor which considers spare capacity of existing systems, existing pipeline protection status and upgradation of existing system

The Project Proposal shall identify all locations where the new pipeline will be mechanically connected to existing facilities with or without electrical isolation and additional cathodic protection requirements shall be assessed accordingly.

Fundamental design calculations shall include:

- a. Determination of the total current required.
- b. Optimum placement and sizing of the anode beds and CP power supplies with consideration given to maintaining potentials on the pipeline within the criteria detailed in Table 4.
- c. For calculations, assume the following:
 - A native state potential of -600 millivolts (CSE), and
 - A pipeline coating conductivity for as listed below should be used for design calculations for new coated pipelines unless the actual coating conductivity measurements values and/or site conditions negate the use of these values.
- d. The design shall include a current and potential attenuation model that considers:

Permanent ICCP systems operating at 75% of rated current output.

Minimum and maximum instant off Potential values of 850 and 1,150 (-mV) Permanent galvanic anodes installed on the pipeline at locations such as road crossings or thrust anchors. For these calculations, magnesium anodes shall be considered to have an open circuit potential of -1.7 volts. Unless field data verifies otherwise, the soil resistivity for the calculation of the resistance to ground of the permanent magnesium anode(s) shall be assumed to be:

- 10,000 ohm-cm in normally dry areas
- 3,000 ohm-cm in areas where the backfill sand is likely to contain small amounts of moisture, and
- 300 ohm-cm in normally wet areas including subkah.
- Bonds to existing pipelines using the measured potential on the existing pipeline and an assumed resistance to ground of 0.015 ohms for the existing pipeline unless otherwise measured.

In the case of new pipelines laid as a stand-alone pipeline, pipeline attenuation calculation shall be carried out considering following method (For calculation details and formulas, refer to Annex E, ISO 15589-1:2015).

- Consider pipeline diameter, wall thickness and length
- Consider maximum and minimum polarized potential values from criteria section
- Consider soil resistivity, coating type
- Consider specific coating conductivity from Table 1
- Consider pipeline native potentials
- Calculate pipeline linear resistance, current, attenuation constant, characteristic resistance, change in potential at drain point, change in potential at the farthest point
- Distance between CP stations

For existing pipeline corridor CP system span data in the form of a table specifying the maximum allowable distances between CP systems based on the existing CP system operational data shall be provided as an input. See Table 2 for typical span.

In cases where either simple attenuation as described above or span cannot be provided, then complete attenuation calculations should be provided considering pipeline coating quality, soil resistivity, existing pipeline potentials, condition of existing and new pipelines.

6.3 Field Data

6.3.1 Project Proposal

The project proposal does not require measurement of field data. The field data including existing CP system details, where available can be provided as an input by the PMT team by collecting the same from Proponent.

6.3.2 Detailed Design

The following field data are required to complete a detailed design package for a cathodic protection system for a new pipeline, and shall be included with all detailed design packages submitted for review (spread sheet for field data shall be submitted in addition to the standard documentation).

a. Pipe-to-Soil Potentials

Pipe-to-soil potentials shall be measured along the proposed pipeline route on all buried pipelines that parallel within 50 meters of the proposed pipeline or cross the proposed pipeline route.

b. Existing CP Power Supplies

The location and equipment designation for any existing CP power supply within 250 meters of the proposed route for the new pipeline shall be recorded. If practical, the rating and operating output of these rectifiers shall also be recorded. If the rectifier ratings and operating outputs cannot be ascertained in the field, a list of these rectifiers shall be submitted by the PMT to the appropriate Proponent organization for determination of the ratings and updated operating outputs.

c. Soil Resistivity

Soil resistivity (conductivity) measurements shall be taken at the locations listed below.

- Along the pipeline route at 1 km intervals to cover the proposed pipeline depth. Such data must include soil conditions such as dry, Subkha, road crossings, pipe crossings, valves, fences, HV crossings, and parallelism.
- Proposed locations for impressed current anodes.
- Proposed locations for galvanic anodes.

Note:

Soil resistivity data along the pipeline corridor may be available either from previous projects or from Proponent. In such cases when the data is already available, then a separate survey along the pipeline corridor is not required.

6.3.3 Soil Resistivity Measurement

6.3.3.1 Surface Anode Beds

- a. Conduct soil resistivity or soil conductivity measurements at 10-meter intervals over the full length of the proposed surface anode bed location.
- b. The four pin Wenner method may be used in areas with nominal soil resistivities below 5,000 ohm-cm. Other non-contact electromagnetic soil resistivity/conductivity instrument shall be used in areas where the nominal soil resistivity exceeds 5,000 ohm-cm.
- c. Measurements shall be taken for the planned installation depth of the anode bed.
- d. Mark the selected location of the surface anode bed with wooden or metal stakes at each end of the proposed anode bed location. Show the location on a detailed site sketch, and include this information on the CP design drawings.

6.3.3.2 Deep Anode Beds

- a. Conduct soil resistivity or soil conductivity measurements for deep anode bed locations.
- b. Mark the selected location of the deep anode bed with a wooden or metal stake. Show the location on a detailed site sketch, and include this information on the CP design drawings.

6.3.3.3 Distributed Anode Beds

- a. Conduct soil resistivity or soil conductivity measurements over a representative sample of the distributed anode installation locations.
- b. Ensure that the measurements are representative for the planned installation depths of the anodes based on the finished grade levels at the site.
- c. Show the location that each measurement was taken on a detailed CP layout drawing.

6.3.3.4 Galvanic Anode Locations

Measure the soil resistivity at the approximate location and depth for the anodes. This resistivity data shall be used in the detailed design to determine approximate anode discharge levels and the effect of the galvanic anode on the attenuation of impressed current on the new pipeline.

6.4 Production Pipelines - Special Considerations

- 6.4.1** The cathodic protection of buried "production" pipelines (i.e., flow-lines, test-lines, water injection lines and trunk-lines, or other pipelines connected directly to a well casing) shall be provided by the CP systems for the well casing(s) in accordance with SAES-X-700 associated with the respective production pipeline for pipeline lengths that are 15 km and lower.
- 6.4.2** For production pipelines that extend more than 15 kilometers from the nearest well casings, CP system shall be addressed separately as a cross country pipeline in accordance with this standard.
- 6.4.3** Production pipelines shall be provided with test stations, bond stations and bonding in accordance with this standard.

6.5 Impressed Current Cathodic Protection

Cathodically protect buried pipelines covered by the scope of this standard with impressed current cathodic protection (ICCP) within 30 days of burial of any section of the pipeline. If this schedule cannot be met, provide temporary cathodic protection in compliance with this Standard.

Exceptions:

Galvanic anodes with a calculated life greater than 15 years may be used instead of ICCP for buried sections of pipelines that are:

reliably electrically isolated from all other underground metallic structures and require less than 2 amperes of current, or are

normally above-ground but have short buried sections such as camel crossings or road crossings.

6.6 Galvanic Cathodic Protection

Unless otherwise noted in this Standard, galvanic anodes are supplemental to the ICCP system and shall be installed according to Standard Drawings AA-036352 and AA-036069 for the structures detailed in Table 3.

6.7 Temporary Cathodic Protection

6.7.1 Temporary CP shall be installed to provide short term cathodic protection only when:

- a. The permanent ICCP will not be energized within 30 days of burial of any portion of the pipeline, and
- b. The magnesium anodes and cross bonds installed during pipeline construction for permanent cathodic protection requirements will not provide enough CP (to achieve temporary criteria) for the new pipeline or pipeline segment.

Note:

The current density requirements and protection levels required for temporary CP can be found in Table 4 of this standard.

6.7.2 The design life of temporary cathodic protection shall be 2 years minimum. However, if for some reason at the end of 2-year period the permanent CP system cannot be installed and/or commissioned, life of temporary CP system may be extended after obtaining a specific return approval for such life extension from Proponent. Design calculations for additional temporary anodes along with pipeline protection status for 2-year period shall be submitted to Saudi Aramco for review and approval.

6.7.3 The design for temporary cathodic protection shall consider the effect of the CP current that will be provided through the bonds that are going to be installed to parallel and crossing pipelines, and galvanic anodes installed at road crossing and thrust anchors.

Whilst bonding of all the pipelines in the corridor is a requirement (see Section 6.10), provision for bonding in the form of pipe to cable connection and cable termination at the bond box shall be considered in the design and installation. However, decision to actually carryout bonding or when to bond is to be made by Proponent taking into consideration the protection status of existing pipeline, impact of bonding to the new and the old pipelines, etc.

6.7.4 Where supplemental galvanic anodes are determined to be necessary for temporary cathodic protection, wherever practical, they shall be installed at permanent galvanic anode locations. The permanent junction box or test station facilities shall be used to terminate the supplemental galvanic anodes.

6.7.5 Pipe-to soil potential survey data verifying that the temporary system is providing adequate protection shall be provided by PMT and submitted to the CP proponent monthly for the first 6 months, and every 6 months thereafter till the commissioning of the impressed current system.

6.7.6 At locations where the permanent galvanic anode equipment will not work for effective temporary cathodic protection:

- Select low resistivity locations to maximize the current output and minimize the number of anodes required.
- Install galvanic anodes on an "as required" basis, i.e., use the calculated numbers of anodes as a guide, and install anodes as required based on meeting the temporary CP protection criteria (Table 4).

6.7.7 Terminate galvanic anodes for temporary cathodic protection in a 3-pin test station using the installation details given in Standard Drawing AA-036352. The number of anodes required per location shall be based on the actual requirement and not as shown on the standard drawing. If the location for the 3-pin test station location coincides with a KM marker/CP test station location, a 1-pin test station is not required.

6.8 Using Existing Cathodic Protection Capacity for New Pipelines

6.8.1 The design of a new cathodic protection system for a pipeline shall take into consideration all aspects of existing cathodic protection that may affect the new pipeline.

6.8.2 Prior to beginning the detailed design for an impressed current CP system for a new pipeline or section of a new pipeline, the Design Agency shall determine the protection levels (pipe to soil

potentials) of close by pipelines (within 50 meters) that may be influenced by the same cathodic protection system(s).

6.8.3 The output levels and capacities of the existing CP systems shall be determined by the Design Agency in cooperation with the CP Proponent.

6.8.4 The Design Agency should consider the use of surplus capacity from the existing CP systems for the protection of new pipelines or pipeline segments. Designs for permanent ICCP systems that propose the use of surplus capacity from existing systems require written concurrence from the CP Proponent Organization. Proper agreement to this effect must be made at the proposal stage (also refer to sections 5.4 & 6.2).

Note:

Proponent or CSD signatures on review packages do not satisfy or substitute for the requirement for written concurrence from the CP Proponent Organization for the use of surplus CP capacity from the CP Proponent Organizations' existing CP facilities.

6.8.5 Where new pipelines share a common corridor with existing pipelines and will be bonded to the new pipelines, upgrading of the existing CP systems may be more cost effective than adding new CP systems in the parallel pipeline corridor. The Design Agency shall specifically address this point in the project proposal design package and include a statement confirming that this has been discussed with the CP Proponent, noting the final agreement on this subject.

6.8.6 Where new pipelines are being added to an existing CP system that is more than 5 years old, the contribution of the existing anode bed to the overall anode bed life and current capacity for the upgraded CP system shall be calculated on a prorated basis using actual utilized life at full rated output as full life. The existing anode bed shall be bonded to the new anode bed when practical and in compliance with this standard.

Exception:

Do not use this method, when the existing CP system is more than 20 years old. Consider completely a new anode bed in such cases.

6.8.7 Where a new pipeline is being added to an existing CP system that is less than 5 years old, an anode bed upgrade is NOT required if the existing T/R and anode bed are capable of providing the required current at an output voltage not greater than 90% of the power supply full rated voltage. If the required current for the existing pipelines and the new pipelines cannot be achieved at 90% or less of full rated voltage, an anode bed upgrade shall be designed to comply on a prorated basis (see 6.8.6).

6.9 Electrical Isolation

6.9.1 Electrical isolation is mandatory between new pipelines and all equipment and facilities operated by a different Proponent Organization. Typical locations where a change in Proponent may occur are:

- a. Onshore-offshore transitions
- b. Pipeline to Plant transitions

Exception:

If commissioning potentials are inadequate due to shorted conditions that cannot be corrected cost effectively, supplemental distributed ICCP or alternative protection shall be provided in lieu of electrical isolation to ensure that all shorted sections of the structure meets protection requirements.

For hazardous locations where if use of isolation devices is prevented for safety reasons, then supplemental cathodic protection shall be provided at these locations to ensure proper protection.

6.9.2 Electrical isolation or supplemental cathodic protection is mandatory at the following locations:

- a. Scraper traps and valve stations

- b. Product pipeline to bulk plant transitions
- c. Pipelines between GOSPs, WIPs or other production facilities operated by a common Proponent.
- d. Grounded lateral pipelines, metallic supports including above ground bypasses and crossovers
- e. Motor-operated valves not in seawater service

6.9.3 For details of pipeline isolation, isolation devices and test see NACE SP0286 and ISO-15589-1.

Notes:

-Where the installation of electrical isolation is required for the purposes of CP, then all possible piping pathways including the main pipe, branches, instrumentation lines shall all be considered for the purpose of electrical isolation.

-Type of isolation joint is not dictated by this standard and is based on pipeline size, service requirements such as temperature, pressure, chemical resistance, mechanical properties. Use of insulating unions, flange type isolation devices, monolithic isolation devices as well isolation spools are all considered appropriate. However, for wet services where the internal fluid may be electrically conductive, use of isolating spool pieces are considered more appropriate unless it can be evidently demonstrated with the help of internal lining and/or coating that flange type connection can be effective.

6.9.4 Do not install electrical isolating devices in buried or submerged portions of pipelines.

6.9.5 Electrical isolation devices shall be protected from lightening and surges by surge diverters / spark gaps. Such surge diverters must be fully encapsulated and suitable for installation in hazardous areas. Their voltage and current rating shall be based on expected surges and capable of safely discharging the expected surge/lightning currents without any damages and/or breakdown. Surge diverters must be regularly monitored and maintained for their integrity. Any failed surge diverters must be replaced immediately. Cathodic Protection design documents shall provide a detailed section for selection of surge diverters as stated above along with a detailed monitoring and maintenance procedure.

6.10 Bonding

6.10.1 Bonding Requirements for Buried Pipelines

6.10.1.1 Bonding shall be provided at the following locations:

- a. Within 500 meters of the start and finish of parallel pipelines and at every 10 km.
- b. Wherever parallel pipelines converge or diverge within 50 meters.
- c. At buried pipeline crossings. If crossing a corridor, bonds are required for each of the parallel pipelines and crossing pipelines in the corridor, providing the separation between adjacent pipelines in the corridor is less than 50 meters.

Exception:

Unless it can be evidently verified that there is nearby bonding within 1 km distance for all the pipelines in the crossing/parallelism.

- d. At all negative drain points (connection point for the negative cable) for ICCP power supplies for pipelines in parallel within 50 meters.

6.10.1.2 All pipeline bond terminations shall be made above grade bond box or junction box complying with 17-SAMSS-008.

6.10.1.3 Bonding shall be completed according to Standard Drawing [AA-036674](#).

- 6.10.1.4** The minimum bond conductor size shall be 16 mm² (#6 AWG). Bond wire size should be based on the bond current and attenuation. Thicker than 16 mm² cable will be required in some places as per current and attenuation calculations.
- 6.10.1.5** Foreign pipelines bonded to Saudi Aramco pipelines shall be bonded through a variable resistor installed inside the bond box.

6.10.2 Bonding is mandatory at the following locations:

- a. ICCP system negative connection to pipe (drain point):

A bond box / negative junction box shall be installed above or immediately adjacent to the pipeline at the termination location (drain point) for the negative cable from the ICCP supply.

A separate bond cable shall be installed to each buried non-abandoned pipeline located within 50 meters of the drain point bond box and shall be terminated in the drain point bond box / negative junction box. The cable from each pipeline shall be sized to facilitate balanced potentials on the pipelines based on field measurements of the potentials, and resistance to remote earth.

- b. New pipelines buried in an existing corridor:

New pipelines buried in an existing pipeline corridor within 50 meters of an existing pipeline shall be bonded to each of the parallel operating (non-abandoned) pipelines and crossing pipelines in the corridor through a bond box at following locations. Unless it can be evidently verified that there is nearby bonding within 1 km distance for all the pipelines in the crossing/parallelism:

- The beginning and end of a parallel segment.
- ICCP power supply locations.
- 10 km intervals along the parallel segment not already satisfied by at least one of the above two bullets.

Notes:

- 1) If a new pipeline is 20 km long with the ICCP power supply located at the midpoint of the line, bonds to all paralleling pipelines within 50 meters of the new pipeline would be required at each end of the line, and at the ICCP power supply location. This arrangement would comply with bullet three above without need for additional bonding.*
- 2) When new pipelines are buried in an existing pipeline corridor, bonding may affect the protection status of both the new and the old pipelines. There will be locations where the existing pipeline CP system protection status and or excess system capacity will help the proposed pipeline and there will be locations where the protection status of the existing pipeline is considered inadequate and the CP system of the proposed pipeline should consider requirements to improve the existing pipeline protection as well at these locations. It is mandatory to ensure that the protection status of the new and the existing pipelines are well demonstrated as per agreement (see sections 6.2, 6.8.4, and 6.8.5) and accepted. At the discretion of the Proponent, there may be specific situations where the proposed pipelines may have to be commissioned and CP protection status demonstrated before any decision on bonding of the new pipelines to the existing pipelines be made.*

- c. Buried pipeline crossings:

Bonding through bond box (17-SAMSS-008) shall be provided at all locations where buried; in-service, mothballed, or foreign metallic pipelines cross.

Exception:

Bonding is only required between a crossing pipeline and the outer pipelines in multiple pipeline corridor, provided adjacent pipelines in the corridor are not more than 50 meters apart.

6.10.3 Common Camel and Road Crossings

Bond normally above grade parallel pipelines, or a combination of above grade and buried pipelines that are separated by a distance of 50 meters or less at least once every 10 kilometers at the nearest common camel or road crossing, unless it can be evidently verified that there is nearby bonding within 1 km distance of the crossing. Place the bond on one side of the crossing according to Standard Drawing AA-036674.

Exception:

For normally above grade parallel pipelines, if there are no camel or road crossings within a 10km parallel section, (i.e., 10 km with no below grade section), then no bonding is required in this 10 km section. Bonds should be placed at the nearest road or camel crossing on each side of the 10 km above grade section.

6.10.4 Electrically Isolated Inline Spools

Electrically isolated spool joints typically used for instrumentation or other applications shall be bonded around, preferably with a metal bond strap bolted to the outer flange on each end of the spool joint.

6.10.5 Electrically Isolating Devices

Provide bond boxes for all isolating devices installed in a pipeline. Bonding to piping on the non-Saudi Aramco plant side of an isolating device shall be through a variable resistor installed inside the bond box.

Exception:

Bond boxes are not required for isolation devices installed for short sections of buried piping protected by galvanic anodes.

6.11 Cased Pipelines

Casing may affect effectiveness of cathodic protection on the carrier pipelines and hence not desirable. However, various types of casing may be used as in some situations as recommended by other Saudi Aramco specifications and/or guidelines at railway /road crossings, Wadi crossings, and other impact locations to protect the pipelines. Such casings may include metallic coated and concrete casings or bare existing casings.

Isolated Casings

When coated metallic casings or coated concrete casings are used, as these shield cathodic protection of the carrier pipe, it must ensure that carrier pipe is well coated and there should no damage to coating of the carrier pipe inside the casing during installation. Proper electrical isolation of the carrier pipe and casing pipe must be maintained at all times. Further to protect the carrier pipe inside the casing any one or combination of following may be used.

- Non-metallic spacers and proper end seals for the casing to prevent ingress of moisture,
- Filling of the annulus with appropriate dielectric material with adequate long-term corrosion protection properties and/or corrosion inhibitors (refer to AMPP SP0200 & SAES-A-134),
- If filling of the annulus is not feasible, zinc ribbon anodes should be provided inside the annulus to protect the carrier pipe and connected to the carrier pipe through a test station.

When uncoated or bare metallic casings are used, the pipeline shall be electrically insulated from the casing using non-metallic spacers. The casing should be sealed using non-metallic end seals and provided with vent pipes at both ends.

When isolated metallic casings are used, these casings need to be cathodically protected with its own galvanic anodes.

Coated Welded Casings

When coated welded metallic casings are used, the impressed current system that is provided for protection of the carrier pipe should have sufficient capacity to protect the welded casings as well.

Other types

When bare concrete is used either as a casing or as weight coat at unstable areas, it shall be ensured that the underlying coating of the carrier pipe is in good condition and that no part of the reinforced metal of the concrete is in contact with the carrier pipe, (reinforcement and the carrier pipe must be electrically isolated from each other).

Note:

This section does not recommend or prevent use of particular type of casing and/or method and material, etc. It only addresses cathodic protection considerations required, when casings are used as required or recommended by other Saudi Aramco standards and/or guidelines (see SAES-L-460).

Method statements as well as drawings for cased pipelines shall be approved by Saudi Aramco. Refer to NACE SP0200 for some guidelines on Steel-Cased Pipeline Practices.

6.12 Cathodic Protection of existing unprotected Pipelines

When it is desired to cathodically protect existing unprotected structures, considering that these are existing pipelines whose condition is unknown, it is recommended to carry out a current requirement test. Current requirement test in addition to providing current requirement for CP, also verifies the capability of cathodic protection system to protect the structure of interest. Such current requirement tests can be carried out utilizing temporary DC power sources not limited to portable generator and rectifier (including batteries with variable resistors), temporary anode beds (usually made of scrap steel) to determine current required for cathodic protection and probable locations for anode beds.

6.13 Monitoring

Monitoring of cathodic protection systems shall be conducted by the Proponent CP organization in accordance with [SAEP-333](#).

6.13.1 Remote Monitoring

6.13.1.1 Remote monitoring equipment shall be provided for ICCP power supplies used for cross-country pipelines. Material specifications shall be as per 17-SAMSS-018.

6.13.1.2 The remote monitoring equipment shall be constructed to interface with the existing remote monitoring system being used by the respective CP Proponent Department. Interfacing of the new & existing RMU is mandatory to ensure that the RMU data is reliably collected and utilized.

6.13.1.3 PMT shall determine the specifications for the existing remote monitoring system and shall be fully responsible for providing this information to the CP Design Contractor and ICCP power supply vendor.

6.13.1.4 Remote monitoring units shall be provided along the length of pipelines at every 5 km intervals and at critical locations listed below.

- Wadi Areas
- Multiple pipeline parallelism & crossing locations
- Mid Point between two CP stations
- Valve stations

- Isolation locations

6.13.2 Test Stations

6.13.2.1 Provide a test station for measuring pipe-to-soil potential according to Standard Drawing [AB-036907](#) at each:

- a. Kilometer marker of the buried sections of pipelines
- b. Negative connection
- c. Thrust anchor
- d. Insulated cased crossing
- e. Thrust bored road crossing
- f. Paved road crossing
- g. Isolation Joints
- h. Pipeline / Railway / River Crossings
- i. Locations of Coupons
- j. OHPL locations
- k. Wadi crossings
- l. Galvanic Anode Locations.

Exception:

Test stations may not be required when some other form of connection to the pipeline is available such as Junction Box, Pipeline Bond Box, Isolation Bond Box, Anode or Negative Junction Box, etc. Any aboveground feature of a pipeline which facilitates electrical connection to the pipeline can also be used for the purpose of potentials measurement.

6.13.2.2 Provide a four-wire current measurement test stations at every 10 km and at upstream and downstream side of the CP T/Rs. At valve stations, bonding wire should be used for current and hence no separate 4wire test station required. Bond wire size across valve station shall be based on the bond current and attenuation.

Note:

Concrete Test stations and Junction Boxes can also be used in areas where it is expected to be beneficial such as vandalism prone areas, locations of OHPLs. Specific approval for use of concrete junction and test stations shall be obtained in writing from the Proponent prior to actual recommendation of concrete type junction boxes and test stations.

6.13.3 Coupon Test Station

The CP probe or coupon shall be installed approximately 500 mm from the side of the pipe, and 1,000 mm deep (see Figure 1) at every 5 km and critical locations listed in section 6.13.1.4. These coupon test stations shall be combined with Remote Monitoring facilities as per section 6.13.1.4.

6.13.4 Soil Access Test Holes

Install soil access test holes, according to Standard Drawing AB-036907, in concrete and asphalt (paved) areas directly over the top of the pipe to be monitored. In concrete or asphalt areas, install one soil access hole at each of the following locations:

- a. Over the pipe, midway between each pair of distributed anodes.

- b. Over the pipe, in line with each distributed anode (directly across)
- c. One at each test station or riser used as a test station if there is no direct close proximity access to the soil.
- d. Above a pipeline at maximum 50 meter spacing in asphalt or concrete paved areas.

6.13.5 Below Ground to Above Ground Transition Areas

At pipeline transitions through asphalt or concrete, provide 30 cm of unpaved soil around the transition. A soil access test hole is not required at transition points with the 30 cm of clearance.

6.14 Stray Current Corrosion

Stray currents include DC or AC. DC stray currents are from neighboring Cathodic Protection system, DC traction systems, welding operations, etc. AC stray current is usually as a result of Induced AC voltages on the pipeline as a result its proximity to the high voltage power lines.

6.14.1 DC Stray Current Corrosion

DC induced stray current is usually indicated by localized fluctuations to the cathodic protection potentials which could be static or dynamic depending on the source of stray current.

DC stray current can be mitigated by one or several of the following methods:

- Installation of a bonding / drainage station between the two pipeline structures;
- Installation of isolating joint;
- By adjusting the cathodic protection system settings to improve the cathodic protection levels; and
- Localized cathodic protection systems.

Refer to NACE SP0169, Control of External Corrosion on Underground or Submerged Metallic Piping Systems as well as Annexure-B of ISO 15589-1, Petroleum, Petrochemical, and Natural Gas Industries - Cathodic Protection of Pipeline Systems On-land Pipelines.

When DC stray current interference are expected or found, mitigation design and materials requirements shall be submitted to Saudi Aramco for review and approval.

At the time of commissioning of CP system stray current interference tests shall be carried out at pipeline crossings and parallelism locations to confirm that there is no stray current interference. Stray current interference if any found shall be mitigated. These field testing & mitigation for stray current shall be submitted to proponent for acceptance.

6.15 AC Interference and Mitigation

AC interference on a buried pipeline can cause corrosion due to AC current exchange between the pipeline and the surrounding soil. This exchange current depends on several factors listed below.

- AC voltage
- Configuration of the AC power line phase conductors
- Configuration of the earthing conductors
- Distance between the AC power line and the pipeline
- Current flowing the AC power line phase conductors
- Average coating resistance of the pipeline

- Thickness of the coating
- Soil resistivity

Therefore, AC interference modelling shall be carried out and mitigation to reduce the impact of AC interference shall be provided considering the following sections 6.15.1 to 6.15.7. Refer to AMPP SP0177, ISO 18086 & AMPP SP 21424 & BS EN 50443 for AC Interference issues, modelling and mitigation .

6.15.1 Qualitative Testing

Qualitative testing for induced AC voltages shall be completed:

- for new pipelines that parallel (within 100 meters) high voltage overhead AC power lines (69 kV or greater) for a distance of more than 500 meters.
- for new pipelines where the overhead AC power lines cross the pipelines at an angle outside the limits stated in [SAES-B-064](#).
- following the Alternating Current Interference Severity Ranking.

The qualitative testing shall be completed as part of the investigative field work required for the detailed design of the CP system.

6.15.2 Quantitative Study

If qualitative testing indicates that AC voltages on the new pipeline may exceed the safe level requirements specified in SAES-B-064 and the Alternating Current Interference Severity Ranking during steady-state conditions as well as under fault conditions, PMT shall commission a detailed study including software modelling (utilizing approved / recommended software / vendor) to quantify expected induced AC voltages during peak load, and shall implement the studies recommendations for mitigation to reduce the voltages to a safe level.

Note:

Irrespective of measured induced AC voltages, whenever there is deviation to SAES-B-064 and / or Alternating Current Interference Severity Ranking requirements, a detailed AC study shall be conducted.

6.15.3 Alternating Current Interference Severity Ranking

6.15.3.1 Consider the Limits of [SAES-B-064](#)

Assess the limits established by SAES-B-064, section 7.6-Overhead Electrical Powerline Crossings considering the following:

- The horizontal angle of intersection between the pipeline and powerlines operating above 69 kV, as per SAES-B-064
- The minimum horizontal separation between any pipeline and the poles or support structures of overhead powerlines, as per SAES-B-064.
- The induced voltages in pipelines and communications cables shall not exceed twelve (12) volts under normal (i.e., steady-state) conditions, as per SAES-B-064.
- The minimum vertical clearance (measured from grade) of overhead powerline conductors at pipeline crossings shall be as per SAES-B-064.
- Underground electrical cable colocation and crossings, as per SAES-B-064 paragraph 7.8.

In additional to clear limits set forth in SAES-B-064, there are other factors that may affect coating degradation and contribute to AC induced corrosion on pipelines that meet the SAES-B-064 limits.

Therefore, careful consideration should be given to the additional risks. Where a risk is identified the guidelines provided here shall be used for a detailed AC interference study and mitigation.

6.15.3.2 Load Current

The load current of HVAC powerlines that cross or parallel a pipeline should be assessed in conjunction with the limits set by SAES-B-064 and further consideration should be given to Table 12, which provides the severity ranking associated with HVAC load current (this table is reproduced from report "Criteria for Pipelines Co-Existing with Electric Power Lines, Prepared for INGAA Foundation, Inc., Prepared by DNV GL, October 2015, Final Report No. 2015-04")

If the phase current falls within the high to very high severity ranking further investigation shall be conducted for particularly if the crossing angle is close to the limits of the 45° to 135°, within 10% of the horizontal separation distance and / or parallel distance limits allowed for by SAES-B-064.

6.15.3.3 Soil Resistivity

Soil resistivity affects both the magnitude of induced AC and the susceptibility to AC corrosion. The AC corrosion process is a function of the AC current density at a coating holiday, which in turn is dependent on the level of AC voltage on the pipeline and the local spread resistance.

The bulk soil resistivity is a primary factor in the corrosion activity.

[Table 13](#) provides the generalized severity rankings of soil resistivity.

If the soil resistivity at a HVAC powerline crossing falls within the high to very high severity ranking further investigation shall be carried out for particularly if the load current risk identified in Table 11 is high to very high, the crossing angle is close to the limits of 45° to 135°, within 10% of the horizontal separation distance and / or parallel distance limits allowed for by SAES-B-064, 7.6 Overhead Electrical Powerline Crossings.

6.15.4 Field Data

6.15.4.1 Project Proposal

The project proposal does not require measurement of field data. Field data available from existing ACM and CP system may be provided as an input by the PMT team by collecting the same from Proponent. Where data is lacking the scope of work should be developed to clearly define all the necessary inputs.

6.15.4.2 Detailed Design

Detailed design shall provide plan view drawings of the overall powerline and pipeline system, showing lengths and separation distances of all power lines and pipelines. Provide the following information in a suitable format as part of the investigation, software modelling and design:

- a. The pipelines under study.
- b. Pipeline Dimensions including burial depth, the diameter, and the wall thickness of all pipelines.

- c. All parallel or roughly parallel high voltage circuits which may influence the pipeline under investigation.
- d. Horizontal distance of the pipeline under study from the centre of the powerline tower.
- e. Horizontal distance of the pipeline under study from the edge of the powerline tower (e.g., from the outside of the nearest tower leg).
- f. Consideration to be given to laterals and other pipelines feeding or being fed by the pipeline under study to assess their electrical continuity with the pipeline under investigation and the associated AC interference risks.
- g. All exposed structures, such as valve sites, off takes and other such facilities on the pipelines of interest.
- h. All insulating flanges on the pipelines of interest.
- i. All anode beds on the pipelines of interest.
- j. Other pipelines which are parallel to the pipelines under study for significant distances (i.e., on the order of 2 km or more), or which cross them, or which come within 10 m of them.
- k. Adequate consideration to be given to electric substations and generating plants, particularly if their assessed AC zone of interference overlaps or extends over the route of the pipeline under investigation.
- l. Electric substations of both ends of each high voltage circuit shown on the map.
- m. Soil Resistivity Measurements should be taken in addition to regular cathodic protection soil resistivity measurements using frequency-selective equipment and / or the Wenner method at the following locations:
 - All exposed structures: e.g., at all valve sites, flow meters, etc.
 - Locations where one or more power lines deviate away significantly from the pipeline or vice-versa, at phase transposition locations, at power line crossings, and at a maximum of 5km intervals along parallelisms.
 - The minimum number of Soil Resistivity measurements shall be taken at the start, centre and end of a parallelism
 - Locations where the pipeline is within 50 m of power line structures or grounds and is within 500 m of substation and power plant locations.
 - Recommended Wenner four pin spacings in meters are: 0.5, 1, 2, 3, 5, 7, 10, 20, 30, 50, 70, 100 or as deemed necessary. This data shall form the basis of soil layer assessment for AC interference and mitigation and modelling purposes.
- n. Exposed Structures in the following locations require mitigation for safety hazard:
 - Valve sites, off takes, flowmeters and other exposed locations along the pipeline under study or at its extremities. Provide drawings clearly indicating the fence line, the locations and dimensions of gates, the property boundaries (i.e., the maximum extent of any gradient control grid which may be required), the locations and diameters of structures protruding out of the ground.
 - Note that for sites requiring protection, safety considerations often require that gradient control conductors extend at least 1 m beyond the fence line: it is therefore best that the fence line be at least 1 m within the property line so that gradient

control grid conductors do not encroach on adjacent property. Furthermore, a layer of crushed rock may be required to extend outside the grid.

- o. Electrical Properties of the following is required:
 - Coating Resistance. Considering all affected pipelines, both new and old.
 - Galvanic Anode Beds. For each anode bed identified, indicate its physical dimensions, configuration of anodes and how the anodes are interconnected, if known. Remote earth resistances of the anode beds, if known.
 - Bonding locations to all pipelines crossing or in parallel with the pipeline of interest, should be clearly identified on drawings. Provide the size and number of cables used to establish bonding. If resistive bonding is used the details should be provided of the resistance values and wattage
- p. Powerline conductor positions and phasing information required as follows:
 - Provide drawings indicating, throughout the common right-of-way, transmission line routing, conductor heights above ground, and structure positions, for all circuits which run parallel to the exposed line (including exposed line feeds or taps) for even a short distance. Please note that at locations where the power line and exposed line veer apart, it is important to know the relative positions of the power line and exposed pipeline up to a reasonable distance apart to ensure accurate modelling of the AC interference and mitigation measures required.
 - Phase Transpositions. Specify the locations of all phase transpositions as well as the configuration of all conductors whose positions change due to the transpositions: gather enough information to allow three-dimensional modelling of all conductors throughout each transposition.
 - Power Line Cross Sections. Provide typical cross sections of the power line right-of-way with:
 - Phases and circuits labelled.
 - Conductor spacings and heights above ground (at structure and mid span) clearly indicated. Include overhead ground/shield wires.
 - Provide conductor type of all phase wires and static wires of all circuits in the common right-of-way. For long counterpoises, please also indicate the conductor type.
 - Frequency 50Hz, 60Hz or other.
 - Remote Substation Locations. For all circuits, specify the distance between the point where they leave the common right-of-way and the substation beyond the common right-of-way to which they are connected. Gather information on the number of powerline towers between the point of departure from the common right-of-way and the substation.
 - Structure Grounding. For all circuits, indicate the footing type of the supporting structures provide a detailed description of the footing dimensions and of any associated grounding conductors, including rebars in concrete foundations. Provide the distance between the structure footings if more than one is associated with a given structure (e.g., for lattice towers).

- Guy Wire Anchors. Indicate at what locations there are guy wire anchors that are electrically continuous with static wires or neutral conductors: i.e., guy wires do not have insulating breakers in them and are bonded to static wires or neutral conductors at pole top.
 - Counterpoises. Indicate positioning, length and conductor diameter of all existing buried counterpoises or other conductor networks which are within or close to the common right-of-way.
 - Nearby Substation and Power Plant Grounding Systems. Provide drawings and / or dimensions of the grounding systems of any substations or power plants which are fed by the respective powerlines.
 - Provide single line diagrams showing networks associated with all circuits present in the common right-of-way.
- q. Ground Resistances require attention as follows:
- Structure Ground Resistances. Provide a listing of the ground resistances of all power line structures in the common right-of-way and of all structures as provided by the powerline operator. If a listing of measured ground resistance values is unavailable, provide typical value(s).
 - Substation Ground Resistances. Provide a listing of measured ground resistance values of all substations to which the circuits existing within the common corridor are connected.
 - Static Wire Connections. For all circuits, indicate whether the static or neutral wires are electrically continuous with the structures which support them and with the substations at which they terminate. Provide the Basic Insulation Level rating for static wire insulators (if any). Indicate which structures are bonded to the static wires and the locations of static wire discontinuities.
- r. Fault Current data to be provided as follows:
- Fault Current Contributions. Provide the single-phase-to-ground fault current contributions (magnitude and angle) from both sides of the faulted circuit and in all other transmission lines or circuits in the common right-of-way, even those whose parallel exposure to the exposed line is short or is far from the fault location, and in all circuits connected to substations/plants fed by the exposed line, for the following fault locations: At roughly 10% intervals of the common right-of-way length on each of the circuits present in the common right-of-way. For circuits with only a short parallel exposure to the exposed line, a minimum of one fault is required. Include data for a fault occurring at any substations which exist within the common right-of-way. If possible, provide currents in all phases (including non-faulted phases), in amps.
 - Soil Resistivity. Indicate what soil resistivity was assumed for the computation of the line parameters used to calculate fault current values.
 - Future Expansion. Indicate for what period the fault current values are valid and what percentage increases can be expected in the future.

- Fault Duration: primary and secondary fault clearing times for all circuits for which fault current data has been provided. Please also provide details on automatic re-closure, if any.
- X/R Ratio. Please provide sub-transient X/R ratio for each fault.
- Fault Type. Were fault current contributions determined? If not, please explain.
- s. Load Current Data. For all circuits within the common right-of-way, even those with only a short parallel exposure to the exposed line provide the following:
 - Phase-to-phase energization voltage.
 - Magnitude and angle of present and well-defined future peak load and emergency load flow. Indicate if maximum load currents in one circuit necessarily coincide with small (or large) load currents occurring in another circuit or circuits: i.e., is there any correlation between the magnitudes and angles of the load currents in different circuits?
 - What percentage increase can be expected to occur in the future for the values provided?
 - Maximum load unbalance for each circuit.
 - Maximum load harmonic currents for each circuit, if applicable.

6.15.5 AC Corrosion

From the AC voltages measured on the pipeline, AC current density shall be estimated utilizing below formula.

$$i_{ac} = 8 V_{ac} / \rho \pi d$$

where

i_{ac} – Calculated AC Current density

V_{ac} – Measured AC voltage on the pipeline near the powerline

ρ – Soil Resistivity around the pipeline where the AC voltages were measured

d – Holiday diameter (use standard value of 1.12 cm unless experience indicates otherwise)

From the above calculations, if i_{ac} is found to be higher than 20 amps/m², then suitable mitigation measures must be carried out to mitigate AC potential and AC corrosion by one or combination of below recommendation.

- Zinc grounding wires/electrodes
- Copper grounding with DC isolation
- Polarization Cell / DC decoupling devices
- Gradient control mats

Refer to NACE standard - SP0177 - Mitigation of Alternating Current and Lightning Effects on Metallic Structures and Corrosion Control.

When AC stray current interferences are expected or found, mitigation design and materials requirements shall be submitted to Saudi Aramco for review and approval.

AC corrosion current density values may exceed the acceptable limit of 20 amps/m² even though the AC voltages are within the permitted limits of [SAES-B-064](#).

Therefore, prevailing soil conditions shall be considered to calculate permitted V_{ac} to limit the Calculated AC Current density to lower than 20 amps/m². Then, lower of these voltages resulting from this calculated induced voltage limit and [SAES-B-064](#) requirement shall be considered to carry out further modelling and mitigation recommendations.

The accelerated corrosion of zinc grounding wires and electrodes has been established through research and result careful consideration shall be given to the AC current drained to ground to ensure that it does not exceed the limits set out in section 8.7 DC Decouplers of this standard.

If cost-effectiveness of the AC mitigation is proven to exceed reasonable engineering expectations the following ongoing monitoring and risk assessment methods shall be included in the AC Mitigation design for review and approval by CSD:

- Electrical Resistance Probe test stations complete with suitable probes and monitoring facilities as per Appendix, Electrical Resistance Soil Corrosion Probes set out in [SAEP-333](#).
- Consideration shall be given to ISO 18086 Annex B.3, NACE SP 0104 paragraph 8.14.3 and ISO 22426 Annex A.3. One such station per 5 km shall be included in the design.

6.15.6 AC Corrosion Coupon & Remote Monitoring

AC corrosion coupons capable of measuring CP ON & OFF Potentials, CP Current Density, Corrosion Rate, AC Induced Voltages, and AC Current Density shall be provided at every 2 km or at select locations of pipelines close to the overhead powerlines as determined from detailed AC interference and mitigation design process. Consideration shall be given to ISO 18086 Annex B, NACE SP 0104, and ISO 22426.

6.15.7 Buried High Voltage AC Cables

Where buried high voltage AC cables either cross or are collocated with a new pipeline, certain considerations must be included in both the Cathodic Protection and AC Interference and mitigation designs. These considerations are as follows:

- At High Voltage AC cable / pipeline crossings the separation distance shall be determined by [SAES-B-064](#) paragraph 7.8 Underground Electrical and Communications Cable Crossings.
- Careful consideration shall be given to pre-existing buried electrical and communications cables and associated utilities are encountered as a new pipeline corridor is being created. Where new cables are to cross a pipeline corridor either duct banks or High-Density Polyethylene (HDPE) pipe is allowed for.
- Where High Voltage AC cables exceeding 69kV parallel a new pipeline and a minimum separation distance of 5m cannot be maintained, then parallel AC mitigation earthing, zinc

ribbon or equivalent, shall designed and installed between the pipeline and HVAC cable for the full length of the collocation with DC decouplers as necessary.

- In instances where AC mitigation earthing is planned due to high voltage overhead powerlines then the effort does not have to be duplicated, but the earthing must be installed between the pipeline and the buried HVAC cable as protection against possible fault current conditions.

6.15.8 Evaluation of AC corrosion likelihood and Monitoring

Evaluation of AC corrosion likelihood shall be carried out considering the following parameters for underground pipelines closer to the AC power lines and prone to AC interference. Facilities to collect and capture these data shall be provided on pipelines closer to power lines (refer to section 6.15.5)

- AC voltage on the structure
- CP ON potential
- IR Free potential
- AC current density
- DC current density
- AC/DC current density ratio
- Soil resistivity
- Corrosion rate
- Spread Resistance

Refer to ISO 18086 & NACE SP 21424 for appropriate acceptable values that should be aimed for whilst assessing these parameters.

7 CP System Performance Criteria

7.1 Cathodic Protection Criteria

Cathodic Protection criteria has been stated in table-4.

7.2 Anode Bed Design Life

7.2.1 The anode bed shall be sized to discharge the CP power source rated current at the anode consumption rate detailed in [Table 5](#), for a minimum of 25 years.

$$\frac{\text{Total Weight of all Anodes (kgs.)}}{\text{Anode Consumption Rates} \times \text{CP Power Source Design Current Capacity}} \geq 25 \text{ years}$$

7.2.2 The minimum design life of galvanic anode systems for electrically isolated short pipeline sections shall be 15 years.

7.2.3 The minimum design life of temporary CP systems shall be 2 years.

7.3 Anode Bed Current Capacity and Consumption Rates

7.3.1 The total current capacity of an anode bed shall be equal to or greater than the design current for the associated CP power source. Impressed current anode design parameters and design data has been included under table-5 & table-6 respectively. Table-7 provides galvanic anode design parameters.

7.4 CP Power Supply Rating

CP systems shall be designed with CP power supplies sized with a minimum rated output current equaling the sum of the current requirement plus a design surplus of 30%.

Note:

When selecting the current rating for power supply units, if the use of 30% surplus results in a rating of say 104 or 105 amps whereas the next standard power supply rating available is 150 amps. It is

economical to use 100 amps which will still give surplus of more than 20%. Therefore, use of surplus capacity must be carefully selected ensuring a minimum surplus of 20.

Standard/Recommended CP Rectifier Rating

For the purpose of onshore pipeline cathodic protection rectifier ratings must be selected from the standard list of rectifiers provided in table-8.

7.5 Circuit Resistance Criteria

7.5.1 The "rated" circuit resistance for an impressed current CP power supply shall be defined as the power supply rated dc voltage divided by the rated dc current. Rated voltages and currents are as detailed on the manufacturer's data sheet/plate.

7.5.2 The "operating" circuit resistance for CP system shall be defined as the total effective resistance seen by the output terminals of the respective rectifier or photovoltaic output control center (OCC).

7.5.3 For a galvanic CP system, the "operating" circuit resistance shall be defined as the open circuit voltage difference between the anodes and the protected pipeline.

7.5.4 For calculation purposes, the operating circuit resistance shall include:

- Anode bed resistance to ground.
- Positive cable resistance from CP power source to anodes.
- Negative cable resistance from CP power source to structure.
- Resistance of the pipeline to remote earth (for calculation purposes shall be 0.055 ohms unless site testing is completed to verify a more accurate value).
- Effective resistance caused by 0.8 volts anode bed back emf (ICCP only) plus the 1.2 volt structure back emf (example: 1.2 volts pipeline + 0.8 volts ICCP anode bed back emf = 2.0 volts total between the anode with coke breeze backfill, and a coated pipeline).

$$R_{emf} = (0.8v + 1.2v) / I_{rated}$$

$$I_{rated} = \text{CP power supply "rated" current}$$

Exception:

CP power supplies with a rated voltage of 12 volts or less shall consider the total back emf to be 1.2 volts for calculation purposes.

Note:

When using R_{emf} to calculate the maximum allowable anode bed resistance to provide 70% of R_{rated} , the value of R_{emf} must be multiplied by 0.7.

$$\text{Example: } R_{anodebed \text{ maximum}} = (0.7 \times (R_{rated} - R_{emf})) - R_{cables} - R_{structure}.$$

7.5.5 New CP systems shall be designed to achieve an "operating" circuit resistance less than or equal to 70% of the CP power supply "rated" circuit resistance.

$$\text{For design: } R_{op} \leq 0.7 \times R_{rated}$$

Where:

R_{op} = CP system "operating" circuit resistance

R_{rated} = CP power supply "rated" circuit resistance

- 7.5.6** The maximum acceptable CP system “operating” circuit resistance measured during commissioning of a new CP system shall be less than or equal to 90% of the CP system “rated” circuit resistance.

Exception:

If a CP system commissions at greater than 90% of the rated circuit resistance, but operating conditions are such that the CP system will be able to provide the required CP for the lesser of 25 years or the predicted life of the structure, supplementation of the anode bed may not be cost effective and deserves additional consideration. In such cases, the requirement for additional anodes shall be at the discretion of the CP proponent organization.

- 7.5.7** If the soil resistivities within a proposed anode bed vary by more than 100%, either additional anodes shall be provided, or, anodes of the same composition with a higher current capacity can be placed in the low resistivity areas so that no anode exceeds the maximum commission current ([Table 5](#)).

7.5.8 Remote Surface Anode Bed

- a. Soil resistivity or soil conductivity measurements shall be taken at 10 meter intervals and at 3 and 6 meter depths in parallel and perpendicular orientations over the full length of the proposed surface anode bed location.
- b. If the average of the measured soil resistivities is above 4,000 ohm-cm, the resistance of each anode, and the combined resistance of all anodes in the anode bed shall be recorded during installation and ensured that the system meets the design requirements before the construction equipment is released from the site.

7.5.9 Remote Deep Anode Bed

Soil resistivity or soil conductivity measurements for deep anode beds are mandatory.

Note:

The intent of 7.5.6 is to eliminate the requirement for “Geonics” soil conductivity surveys in areas that have a low probability of being designed with surface anode beds.

- 7.5.10** Deep anode holes shall be drilled with water or mud solutions in the bore hole. Non-conventional deep anode beds that are drilled with air dry-drilling or foam maybe used in high resistivity fields that normally suffer from cracks, voids and fractures in the formation. Provide steel casings for the active columns of these unconventional anode beds to improve the anode bed performance in loss of circulation zones and include detailed calculations for the steel casing consumption.

- 7.5.11** For deep anode bed designs, drill stem resistance measurements and test anode resistance measurements shall be recorded by the construction contractor on the form contained in Appendix 1 of [SABP-X-003](#) and submitted for review and analysis to the Saudi Aramco (or designate). The drill stem and test anode measurements shall be taken in accordance with the requirements detailed in Standard Drawing [AA-036385](#). In consultation with the Groundwater Division determine the final acceptable borehole depth, and anode distribution.

Exception:

Deep anode holes drilled dry (without water or mud solutions in the bore hole) do not require drill stem and test anode measurements.

8 CP and AC Mitigation System Equipment and Material Requirements

It is mandatory that cathodic protection power supplies, junction boxes, and anodes be pre-approved by Saudi Aramco for cathodic protection material in compliance with the respective SAMSS.

8.1 Anode/Anode Bed General Requirements

- 8.1.1** Impressed current anode materials for soil applications shall be HSCI, MMO, Polymer anodes, etc., complying with [17-SAMSS-007](#).
- 8.1.2** Galvanic anodes for soil applications shall comply with [17-SAMSS-006](#). Design parameters are detailed in [Table 7](#).
- 8.1.3** Zinc ribbon anodes used for AC mitigation applications shall comply with [17-SAMSS-006](#). Detailed design to provide the justification for zinc ribbon dimensions proposed for mitigation use. Refer NACE SP0177 figure 4. The zinc ribbon shall be installed centrally using gypsum (50%) and bentonite (50%) backfill of dimensions 150mm x 150mm for the full length of zinc ribbon. This is to keep the zinc active and to limit the possibility of polarization, that may render the zinc ribbon ineffective as a suitable method of AC mitigation grounding / earthing.
- 8.1.4** Anodes installed deeper than 15 meters are “deep” anode beds, and require drilling depth pre-approval in writing by the Groundwater Protection Division of the Saudi Aramco Reservoir Characterization Department.
- 8.1.5** Spacing between anodes for deep well anode beds shall not exceed the maximum allowable limit stated in the below table-9.
- 8.1.6** Remote anode beds shall maintain the minimum clearances detailed in [Table 10](#) from any other cathodically protected structure.
- 8.1.7** Maintain a minimum separation of 50 meters between adjacent anode beds (deemed remote) powered from different power sources.
- 8.1.8** Install distributed impressed current anodes where a remote anode bed will not provide satisfactory current distribution, or will not be practical.

Note:

Distributed anodes are used for pipelines installed in congested areas, at SSD fence crossings where bare copper grounding wires or bare metallic ground rods are connected to the fence, at full-thrust anchors inside plant areas or fenced areas, at GOSP manifolds or flowline corridors.

- 8.1.9** Unless otherwise specified in this standard, the number, placement, current loading, and length of distributed anodes shall be determined by calculating the soil voltage gradients using the “earth potential rise” equations detailed in Appendix 1 of [SAES-X-600](#).
- 8.1.10** All cable to pipeline connections (test cables, bonding cables, and negative cables) are by thermite welding or pin brazing (refer drawing [AB-036381](#)). All such cable to pipeline connections must be encapsulated/coated according applicable Saudi Aramco coating standard immediately after the connection is cured and ready (within 3 days max from welding) and must be inspected by Saudi Aramco approved coating inspector.

8.2 DC Power Supply

- 8.2.1** Cathodic protection rectifiers and photovoltaic systems shall be manufactured in accordance with [17-SAMSS-004](#) and [17-SAMSS-012](#) respectively and shall be equipped with remote monitoring equipment in accordance with [17-SAMSS-018](#).
- 8.2.2** Install rectifiers and photovoltaic systems in accessible locations, with an all-weather marl road and as far away from sand dunes as possible. Perform sand stabilization, where necessary, to prevent sand dune encroachment.

- 8.2.3** Use of photovoltaic system may be impractical in some areas such as high resistivity soil areas, in such locations rectifiers are preferable.
- 8.2.4** At the location of CP power supply, decision to select Photovoltaic (PV) CP Power Supply installations shall be based on economic evaluation comparing them to conventional power. (Refer to section 5.5 of 17-SAMSS-012).
- 8.2.5** Where AC power is available within 1.5 km distance from the proposed location of CP system, then rectifiers shall be most preferred method of CP power source.
- 8.2.6** Do not use DC power supplies with a rated output voltage greater than 100 volts.
- 8.2.7** Do not install DC power supplies in hazardous locations.

8.3 AC Mitigation and DC Cables

- 8.3.1** AC Mitigation and cathodic protection AC and DC cables shall be manufactured in accordance with [17-SAMSS-017](#).
- 8.3.2** AC mitigation cables sizes selected for installation shall demonstrate suitability as per NACE SP 0177 paragraph 4.8.
- 8.3.3** DC cables connected to a CP power supply either directly or through a junction box shall be optimized in size to compliment the current capacity and resistance requirements of the respective CP system, but shall be #6 (16 mm²) or larger to facilitate the use of stock material.
- 8.3.4** Table 310.15(B) (16) (formerly Table 310.16) (90°C rating column) shall be used to determine the minimum size (allowable ampacity) for HMWPE/HDPE cables. The cable size shall be corrected for an ambient temperature of 40°C.
- 8.3.5** NEC Table 8, Chapter 9, shall be used to determine resistance properties. The resistance of the cables shall be considered in the anode bed design.

Note:

It is not necessary to use High Molecular Weight Polyethylene (HMWPE) cables for negative circuit wiring (test station leads, rectifier to pipeline negative cable, bond wire, etc.). Use of THHN, THWN, and other alternatives noted in 17-SAMSS-017 may be more cost effective and may have less delivery lead time from the vendor.

8.4 Junction and Bond Boxes

Cathodic protection Junction Boxes and Bond Boxes shall be manufactured in accordance with [17-SAMSS-008](#).

8.5 Pipeline CP Assessment Probes

CP assessment probes used for cathodic protection applications on buried pipelines shall be of the same outer dimensions as shown below in [Figure 2](#). The method of construction shown in [Figure 2](#) is intended to be used as typical and is not mandatory and use of other type of CP assessment probes and or coupons are permitted as selected by the proponent. A sample CP assessment probe must be provided to Saudi Aramco for testing and approval before it is deemed acceptable for installation with Saudi Aramco pipelines.

8.6 Grounding

- 8.6.1** Ground conductors shall be jacketed in compliance with [SAES-P-111](#). For locations where, existing bare copper grounding is installed, follow the guidelines given below.
- Insulate buried bare copper conductors crossing a pipeline for a minimum of 6 meters on each side of the pipeline crossing.
The insulation may be PVC conduit, coated cable, or other approved method.

- b. Do NOT use bare copper conductors when running parallel within 3 meters of a buried pipeline or piping.
- c. Do NOT connect a ground cable from the site ground grid to a cathodic protection junction, splice or bond box, nor in any other way establish direct electrical continuity between these boxes and the site electrical ground grid.

8.6.2 Ground rods that are electrically continuous with the pipeline shall not be copper or copper clad.

8.6.3 Where practical, ground rods shall not be installed within 3 meters of a buried pipeline or piping.

8.7 DC Decouplers & Zinc Grounding

8.7.1 ISO 18086:2020 Annex F provides the parameters to consider when choosing a DC decoupling device. Details of the DC decoupler device shall be included in the detailed design documentation for CSD review and approval.

8.7.2 Shall not be installed more than 800 m apart on the same zinc grounding / earthing system.

8.7.3 Shall be installed across isolating and / or monolithic isolating joints where AC interference modelling indicates that AC voltages on the pipeline may exceed 12 V at or near such isolating joints. This is to permit the safe continuous conduction of AC across the isolation joint thereby reducing the risk of unsafe touch potentials.

Note:

In such instances DC decouplers are the preferred method of providing surge protection across the isolating joint and the use of regular surge protection devices shall be avoided as they tend to have higher breakover AC voltage limits that may create unsafe touch potentials in the vicinity of the isolating joints.

8.7.4 Calculations and simulations shall be undertaken during the AC Mitigation design, and the results presented in the final design report, to demonstrate that the net current being grounded through a DC Decoupler to center connected zinc grounding wires meets the maximum design AC current limit per liner meter as per [Table 11](#).

9 Installation, Records, Commissioning, and Inspection

Refer to Saudi Aramco Best Practice SABP-X-003. SABP-X-003 shall be deemed a mandatory document for this standard.

Commissioning shall be in accordance SAEP-332. As built drawings and documents as well as Commissioning Reports shall be submitted to Proponent for review and approval.

Commissioning Reports shall be submitted to CSD for information and feedback.

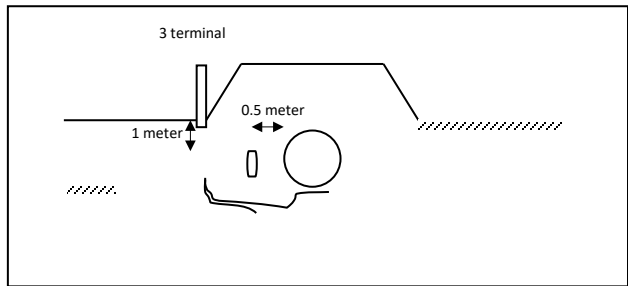


Figure 1 – Typical Coupon Placement

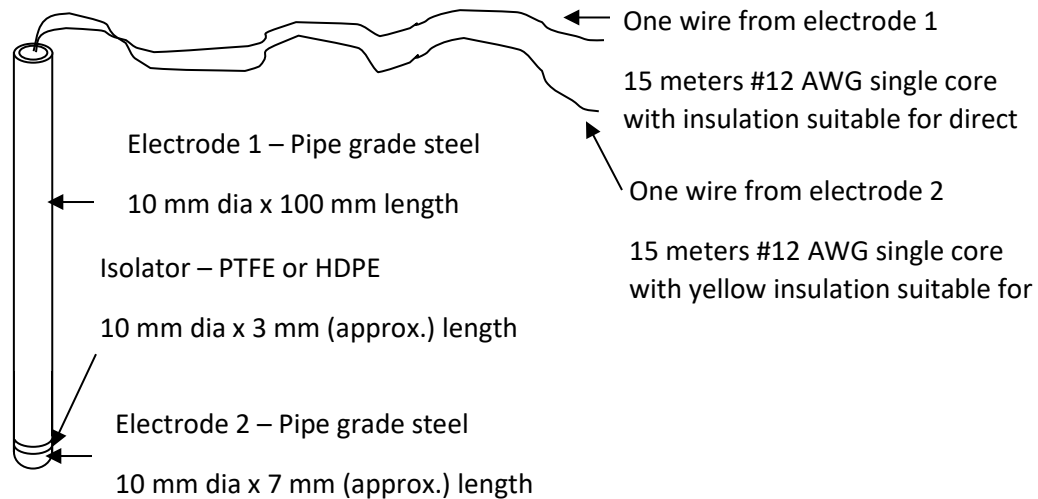


Figure 2 – Saudi Aramco Cathodic Protection Assessment Probe

Table 1 – Specific Coating Leakage Resistance for Attenuation Calculations

Coating Quality Classification (based on coating deterioration)	Soil Classification	Average Specific Coating Resistance ($\Omega\text{-m}^2$)
Excellent	Dunes	3×10^4
Excellent	Damp Sand	1×10^4
Good	Damp Sand (Typical)	5×10^3
Fair	Subkha	1.5×10^3
Poor	Subkha	500
Bare	Subkha	10

Table 2 – Maximum Allowable Distances between CP Systems

Area/Field	Span (km)	
	Recommended	Max. Allowed
Khurais Field	50	75
Northern Area	15	50
Southern Area	30	75
East West	25	50
Central Area	25	50
Shedgum Field	15	50

Table 3 – Magnesium Anode Requirements

Structure Type	Number of Magnesium Anodes ⁽¹⁾	Additional Comment ⁽²⁾
Buried Mechanical Fitting	One 32# prepackaged anode	Connect the anode to the fitting through a galvanic anode test station.
Metallic Casings (i.e., Road or Railway Crossings)	Two 60# prepackaged anodes at each end of the casing for a total of four anodes per casing.	Splice each pair of anode cables together and connect to the casing through a galvanic anode test station at each end of the casing.
SSD fence crossings with bare copper ground cables	Two 60# prepackaged magnesium anodes buried within 3 meters of each side of the bare copper ground cable.	Splice the two anode cables together and connect to the pipe in a single test station located outside the SSD fence at a location convenient for CP Operations.
Thrust bored road crossings or Horizontal Directional Drilled	Two 60# prepackaged anodes at each end of the bore hole, for a total of four anodes per casing/crossing.	Splice each pair of anode cables together and connect to the casing/pipe through a galvanic anode test station at each end of the casing/crossing.
Below grade road crossings and camel crossings on normally above-ground pipelines without dedicated ICCP systems	One 60# prepackaged anode per pipe, plus one additional 60# pre-packaged anode, i.e., one pipe requires two anodes, and two pipes would require three anodes.	For one pipe, place an anode on each side of the pipe along the center line of the crossing. Connect through a galvanic anode test station. For more than one pipe, place an anode on each side of the corridor and an anode in between, or beneath each pipe along the center line of the crossing. Splice the anodes to one cable and terminate in a bond box. Connect a wire directly to each pipeline and terminate each wire inside the bond box.
Pipeline thrust anchors located	Four 60# prepackaged Mg anodes, two on each side of the pipe.	Splice each set of two anode cables together and connect to the pipe

Structure Type	Number of Magnesium Anodes ⁽¹⁾	Additional Comment ⁽²⁾
inside fenced areas		through a galvanic anode test station.

Note ⁽¹⁾ Permanent galvanic anode installations in wet soil conditions or where resistivities are expected to be below 1,000 ohm-cm, shall use bare (not prepackaged) 100 pound magnesium anodes instead of prepackaged 60 or 32 pound magnesium anodes. Where soil resistivity is expected to be below 500 ohm-cm, zinc anodes shall be used.

Note ⁽²⁾ When more than one anode is installed and connected to a test station, splicing the anodes is an option. Alternatively, all the individual anode cables can be run directly to the test station and terminated at the test station.

Exceptions for Table 3:

- 1) Pipeline thrust anchors at well sites with ICCP systems do not require supplemental galvanic anodes.*
- 2) Galvanic anodes are not required for buried flare lines or blow-down lines or associated thrust anchors at well sites if the lines are permanently electrically continuous with the well casing ICCP system.*
- 3) Impressed current anodes may be used instead of galvanic anodes if an existing impressed current distributed anode system is within 100 m.*

Table 4 - Cathodic Protection Criteria for Buried Pipelines ⁽¹⁾

Structure Type	Coating Type	Average Current Density (mA/m ²)	Design Criteria ⁽²⁾			Commissioning Criteria ⁽²⁾					Temporary Criteria ⁽²⁾		
			Minimum "On" Potential (-mV)	Minimum "Off" Potential (-mV)	Maximum Potential (-Volts) ⁷	Minimum Potential (-mV) (must comply with at least one)			Maximum Potential (-Volts) (must comply with at least one)		Average Current Density (mA/m ²)	Minimum "On" Potential (-mV)	Maximum Potential (-Volts) ⁷
						"On" ⁽⁴⁾	"Instant Off" ⁽⁵⁾	"Potential Decay" ⁽⁶⁾	"On" ⁽⁴⁾	"Instant Off" ⁽⁵⁾			
Buried Pipeline (except inside fenced areas)	Not Coated (bare)	20.0	950	850	2.5	900	850	100	2.5	n/a	20.0	850	2.5
	Coated ⁽³⁾	0.10	1,200	850	2.5	1,100	850	100	2.5	1.15	0.005	1,000	2.5
	Coated Production Pipelines	Included with CP system for associated well casing ⁽⁸⁾				1,000	850	100	2.5	1.25	Included with CP system for associated well casing ⁽⁸⁾		
Buried Pipeline Fittings	Coated ⁽³⁾	0.10	1,200	850	2.5	1,000	850	100	2.5	1.15	0.005	1,000	2.5
Buried Pipeline inside fenced areas (valve stations, etc.)	Not Coated (bare)	20.0	950	850	2.5	900	850	100	2.5	n/a	20.0	850	2.5
	Coated ⁽²⁾	0.20	1,200	850	2.5	1,000	850	100	2.5	1.15	0.005	1,000	2.5

- Note 1: Protection criteria stated in here applicable to buried and submerged carbon steel, stainless steel, cast iron, ductile iron materials only.
- Pipelines that are not made of carbon steels, such as stainless steel, duplex stainless steel, etc., will require less negative CP potentials than that of the carbon steel pipelines in accordance with ISO15589-1 & ISO-15589-2. As the protection criteria required for these pipelines are different from that of carbon steel pipelines, proper cathodic protection design consideration shall be made for the same to provide necessary protection whilst limiting the CP potentials to avoid hydrogen induced damages particularly for high strength materials.
 - Similarly, for other materials like copper, Aluminum and other alloys different criteria will be applicable.
- Note 2: All pipe-to-soil potentials measured with reference to a Cu/CuSO₄ reference electrode. Dissimilar metal piping requires that the polarized potential be maintained on all surfaces of the structure based on the protection requirements for the most anodic metal. For pipelines in anaerobic soils in which there are significant quantities of SRBs, the potential values indicated should be increased by further 100 mV. (For example, when using minimum off potential values, it should more negative than -950 mV CSE).
- Note 3: Applies to single layer coated pipelines per [SAES-H-002](#). Current density values of 0.2 mA/m² shall be used for coated areas in Wadi & Sabkha.
- Note 4: The "On" potential is measured with all influencing CP power supplies turned ON.
- Note 5: The "instant off" potential is measured (within 100 milliseconds) after interrupting all influencing CP power supplies. It is recommended this measurement be taken using a permanently installed CP probe or coupon (approx. 10 mm dia. x 110 mm long – see [Figure 2](#)). The probe or coupon should be installed approximately 500 mm from the side of the pipe, and 1,000 mm deep (see [Figure 1](#)).
- Note 6: The "potential decay" is the decrease in potential between the "instant off" and the steady state off potential. A probe or coupon typically reaches a steady state off potential within 60 seconds to 60 minutes. Refer to ISO15589-1 and NACE TM0497 for depolarization measurement procedures. This criterion shall not be used when mixed-metal components or dissimilar metals are in contact for example steel pipeline in contact with copper (pipe or earthing). And in soils containing Sulfate Reducing Bacteria (SRBs), or when interference or stray current activity is present.
- Note 7: Maximum potential limit stated here refers to ON potentials and if with the help of a coupon or other reliable method if it can be demonstrated that the instant OFF values are within the acceptable limits and in such specific cases, higher ON potentials than the limit stated here may be allowed. Further for Wadi & Sabkha area, this maximum allowed potential should not exceed 2000 mV.
- Note 8: With respect to cathodic protection requirements, a production pipeline that extends more than 15 km past the nearest CP power supply shall be treated as a cross-country pipeline. The appropriate attenuation calculations shall be completed and a dedicated CP system shall be provided where required.

Table 5 – Impressed Current Anode Design Parameters

Anode Material	Consumption Rate (kg/Ay)	Maximum Current Density (mA/cm ²)
High Silicon Cast Iron	0.45	0.7

MMO	Not applicable (see manufacturer's specification for 25 year life)	7.0 (fresh water) 35.0 (salt water)
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Table 6 – Impressed Current Anode Design Data

Type	Dimensions	Weight (kg)	Nominal Design Current (Amps)	Maximum Commission Current (Amps)
TA-2 HSCI	56 mm x 2,133 mm	20.9	2.63	4.0
TA-4 HSCI	95 mm x 2,133 mm	38.6	4.45	7.0
TA-5A HSCI	121 mm x 2,133 mm	79.4	5.67	10.0
Tubular MMO	19 mm x 1,200mm	n/a	5.0	7.0
Tubular MMO	16 mm x 1,200mm	n/a	4.2	6.0
Tubular MMO	25 mm x 1,200mm	n/a	6.5	9.2

Notes:

- 1) The consumption rates detailed in [Table 5](#) include utilization factor and effective anode shortening due to end caps.
- 2) The Tubular MMO anodes stated above are typical and other anode dimensions may be selected as required based on the sizes available in the market.

Table 7 – Galvanic Anode Design Parameters

Anode Material	Consumption Rate (kg/Ay)	Potential (mV) (Cu/CuSO4)
Magnesium	10.95	-1700
Zinc	11.8	-1100

Note:

The galvanic anode consumption rates detailed in Table 7 are corrected for efficiency; however, a utilization factor of 0.85 shall be included at the calculations.

Table 8 - Standard/Recommended CP Rectifier Rating

Sl. No.	Output DCV	Output DCA
1	30 V	25 A
2	30 V	35 A
3	30 V	50 A

4	30 V	75 A
5	30 V	100 A
6	30 V	150 A
7	50 V	25 A
8	50 V	35 A
9	50 V	50 A
10	50 V	75 A
11	50 V	100 A
12	50 V	150 A
13	100 V	25 A
14	100 V	35 A
15	100 V	50 A
16	100 V	75 A
17	100 V	100 A
18	100 V	150 A

Note:

When there is a requirement to select or recommend rectifier ratings not stated above, the same shall be approved in writing by Saudi Aramco proponent.

Table 9 - Spacing between Impressed Current Anodes in Meters

Earth Resistivity (ohm-cm)	Carbon Backfill Resistivity (ohm-cm)	
	≤ 10	≤ 50
1,000 and below	3	1.5
Above 1,000 up to 5,000	5	3
Above 5,000 p to 10,000	7	4
Above 10,000	10	5

Note:

Above table on anode spacing is a guideline for anode spacing only and not a recommendation on specification of anode backfill. For carbon coke breeze refer to material data sheet and applicable 9COM.

Table 10 – Minimum Distance to Nearest Buried Metallic Structure

Anode Bed Rated	Minimum Distance in Meters	
	as a Function of Average Soil Resistivity at Anode Bed	

Output Current (Amps)	$\rho < 500 \Omega\text{-cm}$	$500 \Omega\text{-cm} \leq \rho \leq 1,000 \Omega\text{-cm}$	$1,000 \Omega\text{-cm} < \rho \leq 3,000 \Omega\text{-cm}$	$\rho > 3,000 \Omega\text{-cm}$
0 – 35	20 meters	25 meters	50 meters	75 meters
36 – 50	30 meters	35 meters	75 meters	150 meters
51 – 100	65 meters	75 meters	150 meters	250 meters
101 – 150	100 meters	125 meters	225 meters	350 meters

For higher current outputs, the above distances can be pro-rated as 75 meters distance for every 50 amperes of rectifier output.

Notes:

The distances detailed in [Table 10](#) are provided to limit the structure polarized (instant off) potential to less than 1.2 volts and to minimize interference effects on other independent cathodically protected structures. Polarized potentials may be measured using a CP potential probe/coupon, as detailed in [SAEP-333](#) Appendix A-2.

The "Rectifier Rated Output" values apply only to new CP systems being installed by projects. Where a new pipeline is too close to an existing anode bed, the rectifier "normal operating level" should be used with [Table 10](#) to determine the required separation.

Multiple deep anode beds can be treated as individual anode beds if the separation between the anode beds meets or exceeds the minimum distances detailed in [Table 10](#). Example: Two 50 amp deep anode beds placed 75 meters apart in 1,000 ohm-cm can be installed 75 meters away from a buried pipeline.

Table 11 – Maximum Design AC Current Per Linear Meter of Installed Zinc

Zinc Grounding Wire Information	Ribbon Type		
	Standard	Plus	Super
Cross Sectional Dimensions (Inches)	1/2 x 9/16	5/8 x 7/8	1 x 1 1/4
Cross Sectional Dimensions (mm)	12.7 x 14.28	15.88 x 22.22	25.40 x 31.75
Zinc Weight (kg/m)	0.89	1.79	3.57
Wire dia. (mm)	3.30	3.43	4.70
Zn Density (g/cm ³)	7.13	7.13	7.13

Maximum Design AC Current Per Linear Meter of Installed Zinc (A/m)	0.07	0.27	1.05
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Note:

The ribbon type is indicative of commercially available zinc wire anodes. The AC current per linear meter has been derived from the expected AC current induced corrosion on the zinc wire anode over a period of 15 years.

Table 12 – Relative Severity Ranking of HVAC Phase Current

HVAC Current – I (A)	Relative Severity of HVAC Interference
$I \geq 1000$	Very High
$500 < I < 1000$	High
$250 < I < 500$	Medium-High
$100 < I < 250$	Medium
$I < 100$	Low

Note: Above table is a screening guideline only and not a recommendation on the severity of load current which must be included as part of the detailed design with suitable justification.

Table 13 – Relative Ranking of Soil Resistivity (ISO-18086,2020)

Soil Resistivity – ρ (Ohm-cm)	Relative Severity of HVAC Corrosion
$\rho < 2500$	Very High
$2500 < \rho < 10000$	High
$10000 < \rho < 30000$	Medium
$\rho > 30000$	Low

Note: Above table is a screening guideline only and not a recommendation on the severity of soil resistivity which must be included as part of the detailed design with suitable justification.

Appendices

Appendix 1 - Quality Assurance Check List for Pipeline CP System Designs

BI-_____ Pipeline Description _____

"Full Size" Pipeline Plan Drawing with CP Layout

Item	Done	N/A	Pipeline Plan Drawing with all CP Equipment (new and existing)
1	☐	☐	A full size drawing of the pipeline (end-to-end) illustrating placement of all CP equipment, including: ☐ rectifiers, ☐ junction boxes, ☐ bond stations, ☐ test stations, ☐ Mg anodes, ☐ electrical isolation equipment, and ☐ locations where useable AC power is available?
Item	Done	N/A	Details for each <u>Proposed</u> Anode Bed on Drawing
2	☐	☐	The number and type of anodes (TA-4 or TA-5), and configuration, i.e., surface or deep
3	☐	☐	If deep then approved maximum depth from Groundwater Protection Division stated in notes section of drawing
4	☐	☐	☐ Diameter, depth, and spacing of the holes if more than one hole
5	☐	☐	The distance between the pipeline and the proposed anode bed(s)
6	☐	☐	The location of other buried pipelines within 200 meters of the proposed anode bed(s)
Item	Done	N/A	Details for <u>Existing</u> ICCP Systems if Applicable on Drawing
7	☐	☐	The number, type, and age of existing ICCP system anodes
8	☐	☐	Rated and operating, voltage and amperage of existing CP power supplies
9	☐	☐	Existing nearest parallel pipeline (within 200 m) on drawing with low protection areas illustrated

Detailed Drawings of Proposed CP Installations and Electrical Isolation Equipment:

Item	Done	N/A	
10	☐	☐	Isometric sketches of the piping arrangement at the locations where electrical isolation equipment has been proposed and detailed drawings of each ICCP system?
Item	Done	N/A	Details for <u>Proposed</u> DC Power Supplies, Test Stations, Junction Boxes, etc.
11	☐	☐	Rated DC volts and amps
12	☐	☐	Remote monitoring equipment details (compatibility with existing monitoring system)
13	☐	☐	The size and length of all DC cables for the proposed CP system(s)
14	☐	☐	CP junction boxes and bond stations

Calculations

Item	Done	N/A	Calculations for Anode Beds and Miscellaneous Requirements
15	☐	☐	Maximum allowable CP system resistance
16	☐	☐	Maximum allowable anode bed resistance
17	☐	☐	Calculated anode bed resistance based on soil resistivity measurements taken at site
18	☐	☐	Calculations showing CP attenuation characteristics and associated expected pipeline potentials
19	☐	☐	Verification statement that "ALL" buried ground cables electrically continuous with the subject pipeline are insulated (jacketed)

Note: N/A is used in this table as an abbreviation for "not applicable".

Please provide remarks below for any item from above that is applicable but not done:

Item _____

Item _____

Designer Name (Print) _____ Signature and Date _____

PMT Eng. Name (Print) _____ Signature and Date _____

Appendix 2 - Quality Assurance Check List for Pipeline AC Mitigation System Designs

BI-_____ Pipeline Description _____

"Full Size" Pipeline Plan Drawing with AC Mitigation Layout

Item	Done	N/A	Pipeline Plan Drawing with all CP Equipment (new and existing)
1	☐	☐	A full size drawing of the pipeline (end-to-end) illustrating placement of all ACM equipment, including: ☐ DC Decouplers, ☐ junction boxes, ☐ bond stations, ☐ test stations, ☐ electrical grounding / anodes, ☐ electrical isolation equipment, and ☐ nearest powerline tower locations?
Item	Done	N/A	Details for each <u>Proposed</u> DC Decoupler and Earthing System on Drawing
2	☐	☐	The number and type of earthing anodes and configuration, i.e., surface or deep
3	☐	☐	If deep then approved maximum depth from Groundwater Protection Division stated in notes section of drawing
4	☐	☐	☐ Diameter, depth, and spacing of the holes if more than one hole
5	☐	☐	The distance between the pipeline and the proposed earthing anode(s)
6	☐	☐	The location of other buried pipelines within 50 meters of the proposed earthing anode(s)
Item	Done	N/A	Details for <u>Existing</u> GACP Systems if Applicable on Drawing
7	☐	☐	The number, type, and age of existing GACP system anodes
8	☐	☐	Rating and operating, voltage and amperage of existing DC Decouplers
9	☐	☐	Existing nearest parallel powerline (within 200 m) on drawing

Detailed Drawings of Proposed AC Mitigation Installations and Electrical Isolation Equipment:

Item	Done	N/A	
10	☐	☐	Isometric sketches of the piping arrangement at the locations where electrical isolation equipment has been proposed and detailed drawings of each AC Mitigation system
Item	Done	N/A	Details for <u>Proposed</u> DC Decouplers, Earthing, Test Stations, Junction Boxes, etc.
11	☐	☐	Rated surge protection, AC, DC volts and amps
12	☐	☐	At which voltages AC and DC conduction starts and stops including DC blocking voltage
13	☐	☐	The size and length of all cables for the proposed ACM system(s)
14	☐	☐	ACM junction boxes and bond stations

Calculations & Software Simulations

Item	Done	N/A	Calculations for Earthing Anode Beds / Miscellaneous Software Simulation Requirements
15	☐	☐	Maximum allowable system resistance for each DC Decoupler and earthing anode installation
16	☐	☐	Maximum allowable earthing anode bed resistance
17	☐	☐	Calculated earthing anode bed resistance based on soil resistivity measurements taken at site
18	☐	☐	Software fault current simulation output graphs of the pipeline touch potential values and Coating Stress Voltage profiles at all the nearest powerline towers before and after mitigation measures
19	☐	☐	Software induced voltage simulation and predicted AC current density before and after mitigation measures
20	☐	☐	Verification statement that "ALL" buried ground cables electrically continuous with the subject pipeline are insulated (jacketed)

Note: N/A is used in this table as an abbreviation for "not applicable".

Please provide remarks below for any item from above that is applicable but not done:

Item _____

Item _____

Designer Name (Print) _____ Signature and Date _____

PMT Eng. Name (Print) _____ Signature and Date

Document History

2 May 2023	Major	Major periodical revision
02 November 2020	Editorial	Editorial revision to comply with SAEP-301 and SAEP-302
29 March 2019	Editorial	Editorial revision to correct the SAES-X-600 title
29 March 2018	Minor	Minor revision to optionally permit use of isolation devices in hazardous locations
25 January 2017	Major	Major revision to incorporate Value Engineering discussion agreements and further improvement ideas
13 December 2011	Minor	Reaffirmed the content of the document, and reissued with minor revision to incorporate field improvement ideas.